



CONDITION INSPECTION

Vessel inspection - Extended

SURVEY GUIDANCE NOTES **for surveyors**

version 2025.1

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General guidance to the attending surveyor

This survey is performed by BMT Netherlands B.V., Rotterdam ('BMT'), resp. on behalf of CMA Ships as Owners of the vessel ('CMA').

The objective of this inspection is to assess the condition of the vessel, quality of the maintenance as well as the quality of the manager and the associated deployed management systems.

The inspection will be conducted by boarding the vessel and sailing on board for a few days on a selected coastal voyage or during a designated port stay.

Upon boarding, you are requested to introduce yourself to the Master to discuss the scope of the survey. Attention should be given on how best to adjust the survey requirements to the daily routine of the ship and her crew, without disturbing the ship's operations as much as possible. In this respect it will be useful to adjust the order of inspection (what to do first, what next) to the vessel's requirements.



The Inspection

This inspection includes the following tests:

- Starting (one of the) lifeboat engines
- Emergency fire pump
- (One of the) hold bilge alarms, if and when a hold can be entered.
- Operation of the Oily Water Separator and three-way valve by vessel's engineers
- Engine room fire dampers
- Engine room remote stops
- Engine room remote closing valves
- Generator power management system
- Emergency power supply
- All fuel oil and lubrication oil separators
- Fuel oil settling draining
- Onboard LO analysis
- Switching between the primary and secondary electric 440/220 transformers

As set out in this document as from page 9 questions are grouped into sections, which questions can be able to be dealt with more-or-less consecutively. These sections are:

1. Documentation
2. Chief Officer
3. Chief Engineer
4. General
5. LSA/FFE
6. Accommodation
7. Bridge
8. Hull and deck
9. Cargo Gear

10. Engine room

Obviously, an assessment is to be made which sections can be best done at certain times, depending on duties and commitments of the vessel's staff. This is to be discussed with the Master upon commencement of the inspection.

Furthermore, there may be added value in conducting certain parts of the inspection at certain times. For example, it may be useful to be present on the bridge during arrival in port, whereas the engine room is probably best visually inspected whilst at sea and all machinery is in operation.

Some questions are best answered at the end of the attendance on board to get the best possible overview of this item. For example, various questions in the documentation section include an assessment of crew's awareness, which obviously is best done when the crew members have been interviewed and/or observed during their work or when performing tests.

An overview of these questions is as follows:

- 1.4 Requires asking crew members and checking postings
- 1.31 Requires observing crew members in their daily routine
- 1.32 Requires crew's demonstrations of lifesaving appliances
- 1.33 Requires a general assessment of the crew's awareness on procedures
- 1.35 Requires the crew to demonstrate garbage disposal
- 1.41 Requires checking expiry dates on lifesaving equipment
- 1.43 Requires checking expiry dates on firefighting equipment
- 2.10 Requires considering the condition of the outside deck structures

You are requested to obtain a reasonable picture of the condition of the holds, entering as many as accessible. For holds where cargo operations are in progress, appropriate safety measures are to be taken when descending into these holds. If all holds are closed and/or full of cargo, it is normally also possible to inspect at least the hold traverses / cross decks and ventilation arrangements on the forward and aft side of the hold.

A full inspection of all lashing material is obviously not feasible; instead you are requested to inspect, at random, some of the material present on board. This may be equipment in use for securing containers, but also spare stock in storage bins.

When random inspections are done, such as of lashing material, hatch traverses, hatch pontoons, container sockets etc., it is recommended to inspect these items at locations spread out over the vessel's length, so from forward to aft. This will more clearly show the average condition of these items and/or the status of any ongoing maintenance.

Sufficient time is to be spent on reviewing the vessel's on-board maintenance programs, by verifying records and PMS (Planned Maintenance System). The review should include verification that the PMS is set up in accordance the standards laid down by makers and complies with normal engineering practice. It should be verified that the PMS system is working properly and that maintenance is executed timely. Overdue maintenance items should be thoroughly scrutinized and may only be accepted if and when a sound explanation can be provided as why these are overdue or postponed. Under those circumstances, it is required that postponed maintenance

has been approved by the manager.

The scope of survey is not necessarily confined by the list of question. If and when you would come across certain items that are not listed, but in your professional opinion seem to be defective or incorrect, it is strongly encouraged that this is further explored. Defects observed will have to be placed under the scope's most suitable question, with proper explanation given.

It is further also encouraged that additional investigations are made into certain suspicious observations, if needed. For example, if testing of one or two of the hold bilges would prove these to be defective, it is recommended that further investigation into the nature of that failure is made. If required, also additional bilges are to be tested to confirm or exclude conclusions on the cause.

After having completed the inspection, the findings are discussed with the Master and/or any other representative from the vessel's management. The vessel is to be provided with a written list of findings, recommended to be generated by the MATE application (see further 'Reporting'). This list specifically states that it is a preliminary document and that changes may be made afterwards by the administrative staff of BMT during office review.



Reporting

Reporting is to be done using BMT's BizzTouch application. Guidance on the use of BizzTouch can be found in the relevant user guide, which is made available separately.

Part of BizzTouch is a mobile app, which is available for Android and iOS, working on smartphones and tablets/iPads. This app is intended for on-board use to facilitate the inspection, as it works without any active data connection.

The CMA Inspection consists of a number of questions which are to be answered with one of the following options:

- Y (Yes)
- N (No)
- N/A (Not Applicable)
- N/I (Not Inspected)

As from page 9 of this document, each question in the scope is provided with a guidance on how to interpret that question. This will assist you in deciding the correct answer for that question.

For all questions answered with N, N/A or N/I, comments are to be filled in in the respective comment box at the question. This comment should provide more details on the particular defect and its location, the reason why the question may not be applicable or the reason why an item could not be inspected. Please limit this comment to approx. 150 characters. Questions answered with Y, do not require comments, unless these comments would provide essential information.

Please observe that for questions answered with No, the comment will be automatically used to generate the list of findings in BizzTouch. Accordingly, apart from restricting the length to 150 characters, please make sure that the comments form a comprehensible sentence and avoid

telegram style text as much as possible. For example use “Various of the lashing bridges had excessive corrosion, locally wastage” rather than “Lashing bridges corroded”.

If 150 characters is not sufficient, summarize the comments and provide further background in the large comments section at the end of the survey form. This comment section is a free text area where also a summary of the inspection is to be entered. We expect that this summary will be at least one full page of text.

The summary should include details about the circumstances during survey such as the Master having been informed about the survey, were other surveys or bunker operations in progress etc.

Further specific background information on defects is to be stated here as well. This does not have to be a repetition of what was already stated in the comments at the question, but rather a useful addition to that. It could for example entail (verbal) information received on board on intended corrective action or possibly arguments of the crew why they do not agree to particular defects being listed. Furthermore, it is recommended to group similar observations to facilitate reading. For example, various questions answered with No, that comprise corrosion on deck, may be summarized as ‘Main deck structures, railings, hatch pontoons and lashing bridges were corroded.’

The summary could also allow you to share your opinion on items which were not found defective in the sense that the relevant question was answered with ‘No’, but which items may deserve some attention.

All questions answered with ‘N’ (no) should be included in the *List of findings*. When using BizzTouch to generate the list, this will be done automatically.

In case you have doubt as to whether a question should be answered with No, for example because your finding is not specifically described by this survey guidance, you may still decide to list the item on the *List of Findings*. Please choose the most appropriate question to record this observation. You may want to explain to the Master that the final decision on whether to include this observation as a finding, is made during review by BMT. Appropriate comments are required.

Photographs taken during survey, can be uploaded through BizzTouch using the ‘Multi-Upload’ button. First the photos need to be included in a .zip file, then dragged and dropped in the drag and drop field.

Documents can be uploaded, by dragging them in the Media section.

Depending on the condition of the vessel, we require 300 to 400 general photographs, to be uploaded in BizzTouch. It is also requested to rename the Finding photo matching the finding (in case they are not yet allocated to a finding, by using the mobile app). For example, pictures involving question 4.1, could be renamed 4.1 (1), 4.1 (2) etc.

We will prepare the final photo report.

General photographs should show at least cover the following ship areas:

- outside shell plating and hull markings
- main deck and areas for cargo incl. hatch traverses and lashing bridges
- fixed lashing points (sockets, lashing eyes)
- lashing material and lashing applies
- holds, fittings and ventilation
- cargo cranes (exterior and interior)
- mooring decks (forward and aft)
- bridge
- accommodation spaces (galley, mess room, cold stores, corridor)
- engine room, engine control room, steering gear room and emergency generator

General photographs will mostly be overview pictures, but it is recommended to include also some close-ups of the condition of certain items to show their condition, such as for example container sockets, lashing gear, mooring ropes etc.

For deficiencies ('findings'), it is always recommended to include both an overview photograph of that defect, as well as more close-up photographs.

Photographs issued in a format that does not (easily) allow further processing of the pictures, such as a PDF or Word file, is not allowed.

It is not required to sort or re-name photographs.

In case you have any questions on the surveys scope, you can always contact us for information as per following details:

email: risk.surveys@eu.bmt.org / phone (office): +31 10 479 0311.

In case of an emergency outside Rotterdam office hours, you may contact one of the following persons:

Wessel Potjer (+31 6 217 133 43)

Chris Tavernese (+31 6 269 614 44)



Guide to corrosion

Various questions in this inspection involve assessing corrosion and whether that corrosion has reached a stage that it can no longer be accepted. This is not a straightforward task, as corrosion is a developing process and ascertaining the turning point where it reaches unacceptable proportions is subject to interpretation. Essentially, corrosion (or, rust when it involves steel) is an unwanted process and is to be arrested the moment it occurs. However, when this corrosion is limited and superficial, it is not an immediate issue.

We have identified a total of four stages of corrosion, which may assist you in reporting.

Stage 1: superficial

In this stage there is only superficial, often atmospheric rust evident, which can easily be brushed off. The rust development, having a light brown colour, does not affect the strength or functionality of the structure and maintenance can be done by the crew whilst the vessel is in operations.

The question can be answered with YES.



Examples of **superficial** corrosion

Stage 2: progressive

There is development of rust, often of darker colour. Incidental small, loose rust scales are found in way of the object which have parted from the item, but this is limited. Action is required to arrest the corrosion in order to avoid aggravation. Normally maintenance can still be done by the crew as part of the normal maintenance regime.

The question can be answered with YES.



Examples of **progressive** corrosion

Stage 3: **excessive**

There is development of rust which is dark brown discoloured. Loose rust scales are present on / in way of the structure. Maintenance has become more extensive and time consuming and if the rust is spread out over a large area, shore assistance or employment of additional crew may be needed.

The question has to be answered with NO, with the comments stating excessive corrosion.



Examples of **excessive** corrosion

Stage 4: **wastage**

Material is lost, and the functionality of the structure is adversely affected. The structure may be wasted and/or holed and laminar corrosion is evident. Maintenance has become extensive and time consuming and is likely to require shore assistance when spread out over a large area.

The question has to be answered with NO, with the comments stating wastage.



Examples of **wastage**

1. Documentation

1.1 Is the Class status free of outstanding Conditions of Class/Recommendation or, if not, are the Conditions of Class/Recommendations properly followed up?

The inspector should verify if the latest Class status report is free of any Conditions of Class/Recommendations. If the report includes Conditions of Class/Recommendations, it should be verified if proper action is taken for rectification within the specified due date.'

If the records mention (a) Condition(s) of Class/Recommendations without action scheduled for rectification, the answer should be 'no'.

1.2 Is the latest Port State Control inspection free of deficiencies or, if not, are the pending items properly followed up? Please state number of deficiencies in the comments.

The inspector should verify if the latest PSC report is free of any deficiencies. If the report includes deficiencies, it should be verified if proper action is taken for rectification as required by PSC and within the due date specified. If the records show deficiencies which are not properly dealt with, the answer should be 'no'.

1.3 Are DPA details prominently displayed?

The inspector should verify that the identity and contact details of the company's DPA (Designated Person Ashore) are prominently posted and available to all. If details are not posted, the answer should be 'no'.

1.4 Is the crew aware who is the Ship Security Officer and Company Security Officer?

It should be verified as to whether crew members are aware who is the Ship Security Officer and the Company Security Officer. This may be done by asking random crew members and/or checking whether details are conspicuously posted in common areas. If these identities are not properly communicated towards the crew, the answer should be 'no'.

1.5 Are safety meetings held at regular intervals and minutes well documented?

The inspector should verify that safety meetings are held at 4-6 weeks intervals and that the minutes are properly documented. If meetings are not held at the indicated intervals or if the minutes are not properly documented, the answer should be 'no'.

1.6 Is the system for reporting incidents, accidents and near misses being used?

The inspector should verify that the vessel uses an adequate system for recording incidents, accidents and near misses. If no such system is present, or if it is suspected that the reporting system is not functioning, the answer should be 'no'.

1.7 Is the latest external ISM audit free of non-conformities or, if not, are the pending items properly followed up? Please state the number of non-conformities in comments.

The inspector should verify that the latest external ISM audit is free of any non-conformities. If non-conformities are present, it should be verified that corrective action is taken and time frames for rectification are specified. If non-conformities are not

properly dealt with, the answer should be 'no'.

1.8 Are procedures for reporting non-conformities to the office and for receiving feedback adequate?

The inspector should verify that a system is in place to record and report ISM non-conformities, which includes a mechanism to list follow-up actions including time-frames for rectification. It should be verified that communication with the company is effective and that constructive feedback is received. If no reporting system is present, if this system is not properly functioning or if no effective communication with the company's office is established, the answer should be 'no'.

1.9 Is the latest internal ISM audit free of non-conformities or, if not, are the pending items properly followed up? Please state the number of non-conformities in comments.

The inspector should verify if the latest internal ISM audit is free of any non-conformities. If non-conformities are present, it should be verified that corrective action is taken. If non-conformities are not properly dealt with, the answer should be 'no'.

1.10 Is there an organized system in place that shows all certificates and their validity?

The inspector should verify if the vessel maintains a properly organized list of all certificates and that a system is in place to control their validity by marking expiry dates. If control of certificates is being done by the company, it is to be verified that proper communication is in place. If no system is maintained to control the validity of certificates, the answer should be 'no'.

1.11 Is a drug and alcohol policy in place and enforced?

The inspector should verify that a drug and alcohol policy, set out by the company, is present on board. It should be verified that this policy is executed, by verifying records to confirm alcohol and drug tests are being conducted. If no alcohol and drug policy is present or not being enforced, the answer should be 'no'.

1.12 Is there a HSE policy in place and enforced?

The inspector should verify that an HSE (health, safety and environment) policy is present and implemented on board as set out by the company. If no HSE policy is present, the answer should be 'no'.

1.13 Is there a proper system for crew familiarization in place?

The inspector should verify that all crew members joining are given and/or have joined a proper familiarization training. If familiarization is not being done in a proper way, the answer should be 'no'.

1.14 Does the familiarization system ensure that at least a basic safety briefing is given before starting any work?

Although full familiarization may take a couple of days, it should be verified that crew members are given at least basic safety familiarization training before starting any work. If no such basic familiarization is being done, the answer should be 'no'.

1.15 Is the crew aware of emergency procedures?

The inspector should verify that crew members are aware of the various safety and emergency procedures. It should be verified that they are familiar with the different emergency signals and their tasks during the different emergency situations. If crew members are not aware of their duties or procedures, the answer should be 'no'.

1.16 Is there evidence of risk assessments being applied on the work site?

The inspector should verify that risk assessments are being made prior to hazardous work being undertaken, in particular out of routine jobs. This may include hot work, working in enclosed spaces, working aloft, special cargo operations etc. If no risk assessments are being made prior to hazardous work, the answer should be 'no'.

1.17 Is there evidence that a safety permit system is used for relevant types of work (hot work, working in enclosed spaces)?

For any hazardous type of work, the risk assessment should include a permit system, including a checklist with required tasks to be performed prior to, during and after subject work. Permit systems should be complete, properly filled in and signed by all crew members involved. If no permit system is set up or being used, the answer should be 'no'.

1.18 Is there a system in place for storage and handling of dangerous materials?

The inspector should verify that the vessel has a system in place for storage and handling of dangerous materials, such as chemicals. The system should describe where dangerous materials are to be stored, how they should be stored and it should include posting relevant MSDS sheets in way of the storage area. If special handling is required, for example because of the size of the packing, instructions for handling should be included. If no system for storage and handling of dangerous goods is available, or if that system is not properly followed, the answer should be 'no'.

1.19 Are records available indicating that regular reviews are made of the Ship Security Plan?

The inspector should verify that a regular review is being made of the Ship Security Plan and its execution. If records are not available or do not state that regular reviews are made, the answer should be 'no'.

1.20 Are port security procedures in place covering visitors, storing and gangway watch?

The inspector should verify that the vessel maintains procedures for attending visitors, special attention for taking on board stores and adequate schedules for maintaining a gangway watch at all times. If such procedures are not available, the answer should be 'no'.

1.21 Are ISPS procedures for entering the ship followed correctly?

When boarding the vessel, the inspector should verify, through the followed ISPS procedure at the gangway, if the routine is adequate. Appropriate entries should be made in the visitor's log book and a visitor should be asked for a legitimate identification. In case of doubt, the inspector can ask to review the ship's documented procedures. If ISPS procedures are not properly followed, the answer should be 'no'.

1.22 Does the Superintendent visit the vessel at least annually?

The Superintendent's visits should occur at one year intervals. If time interval is not adhered to, the answer should be 'no'.

1.23 Are the deficiencies reported by the superintendent properly followed up? Please state the number of deficiencies under comments.

The inspector should verify if the latest Superintendent visit report is free of any observations. If observations are present, it should be verified that corrective action is taken. The inspector should verify that the superintendent's reports of previous visits are available on board and that follow-up is given to the deficiencies identified in this report. If this is not done, the answer should be 'no'.

1.24 Is there a planned ballast tank inspection regime in place?

Ballast tanks should be inspected at least annually, and records of all inspection results should be maintained. If no records of ballast tank inspections are present, the answer should be 'no'.

1.25 Do recent ballast tank inspection reports show the tanks to be in sound condition?

Records stating the results of all inspections should be maintained. Including details of the tank and any anomalies noted, such as structural deterioration, coating breakdown, leakage of pipework and excessive sediment build-up. If intolerable deficiencies are noted without having been given proper follow-up, the answer should be 'no'.

1.26 Is the Safe Manning Certificate available and signed by the relevant Flag State?

The inspector should verify that the vessel carries a Safe Manning Certificate that is signed by the relevant Flag State. If no properly signed SMC is available, the answer should be 'no'.

1.27 Is the number of crew compliant with the minimum requirements of the Safe Manning Certificate?

The inspector should verify that the number and ranks of the crew members comply with the requirements set out in the Safe Manning Certificate. If the number of crew members is less than required, or if certain required ranks are not on board, the answer should be

'no'.

1.28 Are crew certificates valid and properly endorsed?

The inspector should verify that the crew certificates are valid and appropriate for the corresponding rank, and that the certificates are properly endorsed by the relevant Flag State authority. If crew certificates are not valid or improperly endorsed, the answer should be 'no'.

1.29 Are crew handovers done in an adequate manner?

The inspector should verify that handing over to relieving crew members is done in a proper way. At least crew members in senior ranks should write a proper hand-over document for their reliever. If no proper handing over system is in place, the answer should be 'no'.

1.30 Does the company have a briefing / debriefing policy for Masters and Chief Engineers?

The inspector should verify that the company has a proper system in place for briefing Masters and Chief Engineers before joining a vessel and a debriefing system after their term. If no such system is in place, the answer should be 'no'.

1.31 Are the crew members properly qualified for their duties operating the equipment of the vessel?

The inspector should, through observations and random inspections, verify if the crew members on board seem suitable for their respective tasks. This can be done by observing them in their daily routines or by forming an opinion during the various interviews held. If the inspector is of the opinion that one or more crew members is/are not fully qualified for undertaking (certain) tasks on board to which that crew member(s) is (are) assigned, the answer should be 'no'.

1.32 Is the crew familiar with operating firefighting, life saving and other emergency equipment on board?

The inspector should verify that crew members responsible for the operation of safety equipment, are familiar with that operation. As part of this inspection the crew is asked to demonstrate or operate various kinds of life saving and firefighting equipment, which should enable the inspector to form an opinion on their skills. If the inspector is of the opinion that one or more crew members are not skilled in operating of safety equipment, or if they lack knowledge of the purpose and functioning of that equipment, the answer should be 'no'.

1.33 Is the crew conversant with the company procedures / systems?

The inspector should verify that crew members are generally aware of the company procedures and systems, which is best to assess at the end of the inspection. Obviously not all crew members will be aware of every procedure, so a distinction should be made

between different roles and their responsibilities. If the inspector is of the opinion that one or more crew members is clearly lacking understanding of the company's procedures for as far as these apply to his responsibilities, the answer should be 'no'.

1.34 Is the Master / Chief Engineer aware of financial budgets?

The inspector should verify whether the two senior officers (the Master and the C/E) are aware of financial budgets for the vessel, as set by the management company. This should be reflected by relevant records, communications or similar documents. If the Master and Chief Engineer are not aware of any financial implications of certain actions they may take, the answer should be 'no'.

1.35 Is the crew aware of the environmental regulations?

The inspector should verify whether the crew is generally aware of the environment / environmental hazards and the applicable rules and legislation. This can be done by asking crew members to demonstrate how garbage is disposed of / handled, how waste oil is handled, how pollution is prevented etc. All actions have to apply for the respective roles on board; a cook will be involved in garbage disposal but obviously not in waste oil disposal. If the inspector is of the opinion that one or more crew members is not sufficiently aware of the requirements to mitigate risks for the environment, applicable for his/her tasks, the answer should be 'no'.

1.36 Are the vessel's ballast water management procedures properly followed up?

As per 8 September 2017 all existing vessels need to be compliant with the requirements of the 2004 BWM Convention. Ships must manage their ballast water so that aquatic organisms and pathogens are removed or rendered harmless before the ballast water is released into a new location. This means vessels have to carry an approved water ballast management plan on board, maintain a ballast water record book and have an International Ballast Water Management Certificate. There are two ballast water management standards (D-1 and D-2). The D-1 standard requires ships to exchange their ballast water in open seas, away from coastal areas. The D-2 standard specifies the maximum amount of viable organisms allowed to be discharged. From the date of entry into force of the BWM Convention, all ships must conform to at least the D-1 standard; and all new ships, to the D-2 standard. A ship undergoing a renewal IOPPC after 8 September 2019 will need to meet the D-2 standard by the date of its renewal survey. If a ship does not have an IOPPC renewal survey, then the ship should meet the D-2 standard at a date determined by its flag State, but not later than 8 September 2024. The inspector should verify that a system is in place compliant with the BWM Convention terms and verify that records of ballast water management are properly kept. If not, the answer should be 'no'.

1.37 Is the vessel complying with EU MRV regulations for reporting CO2 emissions?

As per 1 January 2018 the EU MRV system is enforced, which requires ships >5000 gt calling EU ports, to report on CO2 emissions. A monitoring plan for the vessel should have

been submitted in 2017, whereas the 2018 calendar year has been the monitoring phase. The company was obliged to report to the EU committee by 30 April 2019. After 30 June 2019 the vessel should carry a document of compliance from the EU. The inspector should verify that this procedure was followed for the vessel. If not and/or if the vessel does not carry a certificate of compliance, the answer should be 'no'.

1.38 Are the crew's resting hours compliant with the MLC 2006 convention rules?

The inspector should verify, by reviewing crew working hours sheets, that the crew's resting hours are compliant with the MLC 2006 requirements (Maritime Labour Convention). The MLC 2006 general requirements require each crew member to have 10 hours of rest in any 24-hour period, which may be divided into no more than 2 periods, one of which shall be at least 6 hours in length, and no more than 14 hours between any consecutive periods; as well as 77 hours rest in any 7 day period. If resting hours are not compliant, the answer should be 'no'.

1.39 Is the vessel's SEEMP properly documented, including recordings of evaluation measures?

The inspector should verify if a SEEMP (Ship Energy Efficiency Management Plan) is set up and if evaluations of the improvement measures under SEEMP are made and (properly) recorded. As per 1 March 2018 the plan needs to include described methods for data collection (SEEMP Part II), which methods are to be approved by the Flag Administration latest 31 December 2018. These mandatory data recordings should have commenced as per 1 January 2019. If no (properly documented) SEEMP is present or if no confirmation of the Flag Administration is present, or the plan does not include evaluation measures, the answer should be 'no'.

1.40 Is an adequate ship-shore reporting system in place?

The inspector should verify that there is an adequate system in place for communication between the vessel and shore-based offices, in particular the office of the vessel's managers. It should further be verified that there are different systems in place (email, telephone, satellite) to ensure that emergency communication is possible at all times. If no adequate communication system between the vessel and the shore is in place, the answer should be 'no'.

1.41 Are all required inspections of lifesaving equipment up to date?

The inspector should check relevant inspection reports of all life saving equipment, to verify that this equipment has been inspected or serviced by shore-based companies within the time frames set for that equipment. The inspector should, during his inspection of the vessel, at random verify that expiry dates that are stated on the equipment comply with these inspections. If life saving equipment has not been inspected or serviced within the required time frames or is not properly labelled, the answer should be 'no'.

1.42 Are records of lifesaving equipment maintenance available?

Apart from shore-based maintenance for lifesaving equipment, also the crew is required to conduct regular inspections of this material and conduct required maintenance. The inspector is to verify that records show that these inspections are conducted as per time frames set in the vessel's PMS. If no records of this maintenance are available or if the records show that inspection is overdue, the answer should be 'no'.

1.43 Are all required (monthly, annual, bi-annual) inspections of firefighting equipment up to date?

The inspector should check relevant inspection reports of all firefighting equipment, to verify that this equipment has been inspected or serviced by shore-based companies within the time frames set for that equipment. The inspector should, during his inspection of the vessel, at random verify that expiry dates that are stated on the equipment comply with these inspections. If firefighting equipment has not been inspected or serviced within the required time frames or is not properly labelled, the answer should be 'no'.

1.44 Are records of firefighting equipment maintenance available?

Apart from shore-based maintenance for firefighting equipment, also the crew is required to conduct regular inspections of this material and conduct required maintenance. The inspector is to verify that records show that these inspections are conducted as per time frames set in the vessel's PMS. If no records of this maintenance are available or if the records show that inspection is overdue, the answer should be 'no'.

1.45 Are records available to show that regular abandon ship and fire drills have been carried out?

The inspector should verify that records are present that indicate that safety drills have been carried out at required intervals. SOLAS requires every crew member to take part in at least one abandon ship drill and one fire drill every month. If more than 25% of the crew is new, these drills have to be carried out within 24 hours after leaving the port. If drills are not carried out at specified intervals or not properly recorded, the answer should be 'no'.

1.46 Are records available to show that regular SOPEP drills have been carried out?

The inspector should verify that SOPEP based (oil) spill drills are conducted every 3 months, or more frequent if the company procedures require so. Company procedures may further require drills to be conducted prior to each bunkering. If drills are not conducted and/or recorded in accordance with these time frames, the answer should be 'no'.

1.47 Is there a waste / garbage management plan available?

The inspector should verify that a waste and garbage management plan is available on board and that disposal of waste is performed in accordance with this plan. If no adequate waste / garbage plan is present or if disposal is not being arranged in accordance with this plan, the answer should be 'no'.

1.48 Do the vessel's records indicate that there is an adequate preparation made for dry docking?

The inspector should verify if records relating to the vessel's last dry dock show that adequate preparations are being made for docking. This preparation should include a docking plan, giving attention to upcoming repairs and maintenance, obtaining relevant details of previous docking, preparation of plans and drawings, verifying operations of shore connections, consideration for tank quantities and stability and considering spare parts and provisions. If the dry dock preparation is deemed not sufficient, the answer to this question should be 'no'.

1.49 Is the assessment of cyber security performed and the relevant procedures followed up?

IMO has adopted the resolution MSC.428(98) on Maritime Cyber Risk Management in Safety Management System (SMS). The resolution encourages administrations to ensure that cyber risks are appropriately addressed in SMS no later than the first annual verification of the company's Document of Compliance (DoC) after 1 January 2021. This means vessels need to have approved procedures dealing with cyber risk management to:

- identify the roles and responsibilities of users, key personnel, and management both ashore and on board
- identify the systems, assets, data, and capabilities that, if disrupted, could pose risks to the ship's operations and safety
- implement technical and procedural measures to protect against a cyber incident, timely detection of incidents and ensure continuity of operations
- a contingency plan which is regularly exercised.

The inspector should verify that passwords (with 8 characters) are not written in the vicinity of vessel's computers, randomly check that data carriers (USB) are scanned for viruses before use. If not, the answer should be 'no'.

2. Chief Officer

2.1 Are deck and port log books correctly maintained?

The inspector should verify that the deck log book and the port log is filled in correctly. If no separate port log is used, details may be entered in the deck log book. Cargo operations performed are to be stated, as well as statements on extraordinary occurrences, such as damage or delays. Entries that are amended should be marked with initials. The use of correction fluid in official log books is never allowed. If the deck log book and/or port log book are not properly maintained, the answer should be 'no'.

2.2 Is the vessel operated within maximum lashing forces?

The inspector should check the departure conditions as calculated by the ship's loading computer, for some randomly chosen previous voyages. The Chief Officer should be able to indicate how the loading program shows forces and whether these are out of tolerance. The inspector should check whether lashing forces have been exceeded on these voyages, which is normally indicated by a percentage exceeding 100%. Reducing lashing forces can normally be done by applying additional lashings, changing the vessel's stability (GM) or restowing to heavy containers. If the Chief Officer is unable to demonstrate his understanding of the loading program and lashing module or if the program shows that lashing forces were exceeded (values higher than 100%), the answer should be 'no'. If the loading software does not have a lashing module, the answer should also be 'no'.

2.3 Is a lashing inventory made regularly to confirm compliance with the CSM?

Upon newbuilding, a vessel is laid out with a stock of lashing material (twist-locks, lashing bars, turnbuckles etc), which is set out to accommodate every desired loading condition. This stock is defined in the CSM (Cargo Securing Manual) as being the minimum required lashing gear quantities. Normally a vessel is taken on charter on basis of this stock. Not complying with this minimum required stock may result in a vessel not being able to comply with a loading condition required by Charterers. Accordingly, the inspector should check if the lashing inventory is regularly checked to confirm compliance with the minimum quantities of lashing material as per CSM. Slight deviation, up to a few percent from the required stock, should not be considered a problem, provided that this is recorded and properly followed up. Letters from Class stating the position of Class on the stock do not replace this requirement. If a lashing inventory check has not been performed within the last 6 months or if it shows that the quantity of lashing and securing material is significantly less than minimum required as per CSM, the answer should be 'no'.

2.4 Is the cargo gear record book up to date? Please state the last Class annual and 5 yearly inspection dates.

If the vessel is fitted with cargo gear, the inspector should verify if the cargo gear record book is up to date, viz. whether quinquennial and annual inspections by Class have

been performed within the required time frame. Furthermore, intermediate inspections by the vessel's crew should be mentioned. If the cargo gear record book is not up to date, the answer should be 'no'. If the vessel is not fitted with cargo gear, the answer should be 'N/A'.

2.5 Are certificates available for the crane wires and is an installation date and location where the wire is fitted indicated?

If the vessel is fitted with cargo gear, the inspector should request for certificates of the (runner) wires. These certificates, provided when new wires are delivered on board, should indicate the date of manufacturing and the designed SWL. The numbers of the wires listed in the certificates should be compared to the numbers entered in the vessel's records to identify these wires. The crane number and position of the wire is usually not printed on the certificate but may be written on it by hand. Since crane wires age, it is recommended that crane wires are turned after 5 years and renewed after 10 years, but this is not compulsory. If no wire certificates are available, if wires mentioned on the certificates are not identified as wires installed or if the SWL is not in accordance with the SWL of the crane, the answer should be 'no'. If the vessel is not fitted with cargo gear, the answer should be 'N/A'.

2.6 Is the maintenance of the cargo gear up to date?

If the vessel is fitted with cargo gear, the inspector should check the vessel's maintenance records to verify if maintenance of the cargo gear has been conducted within the required time frame and whether any items are overdue for more than 30 days. If the vessel is unable to indicate maintenance of cranes or if maintenance items are overdue, the answer should be 'no'. If the vessel is not fitted with cargo gear, the answer should be 'N/A'.

2.7 Is a planned deck maintenance system in place and being used?

The inspector should verify if a planned deck maintenance system is in place, containing scheduled maintenance for the deck department on basis of time frames. If no planned maintenance system is followed, the answer should be 'no'.

2.8 Is the deck PMS set up in accordance with requirements of makers?

The inspector should, through random checks, verify that the PMS for deck maintenance is compliant with the requirements set out by the maker of the various equipment. If these checks show that the PMS allows longer time frames for maintenance than the makers require, the answer should be 'no'.

2.9 Is less than 5% of all deck maintenance items overdue?

The inspector should check the maintenance records from the deck department. Systems normally can be filtered, showing the total number of items for the deck department. It should be verified that none of the items is overdue for more than 30 days without any explanation. If more than 5% of the items is overdue or items are

overdue for more than 30 days without being explained, the answer should be 'no'.

2.10 Is there evidence of an adequate deck maintenance plan?

The inspector should verify that, next to the recorded maintenance jobs in the PMS, the vessel maintains a plan for maintenance on the external deck structures, including deck, hatch covers and hatch traverses. Especially when external structures exhibit progressing or even excessive corrosion, the Chief Officer should be able to present a plan on how to conduct maintenance to arrest this corrosion by chipping and painting. If no deck maintenance plan is present, the answer should be 'no'. For newly built vessels that have been in service for a few months only with the deck in excellent condition, a deck maintenance plan may not be required and the question could be answered with 'N/A'.

2.11 Are bilge alarms tested at least monthly?

The inspector should verify if (cargo) hold bilge alarms are tested at least once a month and that test results are recorded. If hold bilges are not tested monthly and/or records do not indicate these testing having been conducted, the answer is 'no'. If holds are not fitted with bilge alarms by design, the answer should be 'N/A'. In that case, however, it should be verified that bilges are sounded daily and sounding records are kept. If no records are kept or the records are incomplete, the answer should be 'no'.

2.12 Is the fuel oil sounding system being verified by manual sounding on a monthly basis?

The inspector should verify if a remote tank sounding system for the fuel oil tanks is fitted. The readings of this system should be double checked through manual soundings at least monthly. Sounding results and in particular differences should be recorded and actions taken to correct the deviation. If no manual soundings are being taken for calibration purposes or if manual soundings cannot be taken at all (due to the absence of sounding pipes), the answer should be 'no'. A respective remark is to be inserted in the comments. If no remote tank sounding system is present and tanks are always sounded manually, the answer should be 'N/A'.

2.13 Are maintenance records of the lashing gear available on board?

The inspector should ask the Chief Officer the number of lashing equipment that was overhauled in the last month and ask for records to demonstrate this. Such overhaul involves twist locks (i.e. CV-20, CV-8B) and turnbuckles. The inspector should verify that the maintenance program is followed by a visual inspection of the lashing gear workshop, usually situated in the forecastle space. If no lashing gear maintenance program is followed or if maintenance seems inadequate, the answer should be 'no'.

2.14 Is the ballast and bilge system in an apparent operational condition?

The bilge and ballast system on container ships is often controlled from the cargo office / ship's office. The inspector should check whether the control system appears

operational. In case ballasting operations are performed through a ballast panel, any signs that it may not be operational, such as posted notes with warnings, pieces of tape applied to disable the use of buttons/switches, loose wires etc. are reason for further investigation. If required, an operational test may be requested. Furthermore, it should be verified if any tanks are temporary out of order; this may require further investigation. In case ballasting is performed through a computer controlled system, efforts should be made to assess the operational condition. If the bilge and/or ballast system is not operational or if tanks are out of use without proper explanation, the answer should be 'no'.

2.15 Is stevedore damage properly recorded and communicated and are stevedore damage reports presented to stevedores?

Stevedore damage should be noticed and recorded immediately and a corresponding stevedore damage notice is to be prepared and handed over to the stevedores' representative, obtaining a signed copy in return. The damage notices are to be communicated towards owners and operators in accordance with relevant procedures. If stevedore damage notices are not issued immediately after a damage occurring or are not properly communicated, the answer should be 'no'.

3. Chief Engineer

3.1 Is an engine room PMS in place and being used?

The inspector should verify if a planned engine room maintenance system is in place, that contains scheduled maintenance for the engine room department on basis of time frames and/or running hours. If no planned maintenance system is followed, the answer should be 'no'.

3.2 Is the engine room PMS system apparently adequate in its operation and record keeping?

There is no strict requirement for a PMS to be computer-based, although this is recommended. Any PMS should adequately translate maintenance requirements into actions and provide insight on pending and overdue jobs. If the PMS is maintained in hard copy or possibly through a library of files (Word, Excel), it is to be verified that the system works efficiently and is reasonable protected against errors or malicious intentions. If the PMS system does not appear to function as expected, the answer is 'no'.

3.3 Are maintenance intervals in the engine room PMS in accordance with requirements of the makers?

The inspector should, through random checks, verify that the engine room PMS for engine room maintenance is compliant with requirements set out by the maker of the various equipment. If these checks show that the PMS allows longer time frames for maintenance than the makers require, the answer should be 'no'.

3.4 Is less than 5% of all engine room maintenance items overdue?

The inspector should check the maintenance records from the engine room department. Systems normally can be filtered, showing the total number of items for the engine room department. It should be verified that none of the items is overdue for more than 30 days without any plausible explanation. If more than 5% of the items is overdue or items are overdue for more than 30 days without being explained, the answer should be 'no'.

3.5 If items are overdue, can a reasonable explanation be provided and is the risk of failure mitigated?

Sometimes maintenance items that are overdue involve deferred maintenance. If that is so, the Chief Engineer should be able to demonstrate that this postponement is done in a carefully and thoughtful manner. At least approval from the vessel's technical management is to be obtained, but it is recommended that any decision to defer maintenance is backed up by makers of the equipment. If maintenance is deferred due to the lack of spare parts, the Chief Engineer should be able to explain why spare parts could not be delivered in time. The Chief Engineer should be able to show that these parts have been ordered and will arrive shortly. Maintenance for engines sometimes may be deferred if the makers apply condition based maintenance (CBM) which often

cannot be properly accommodated in the PMS; deferral because of CBM should also be supported by documents / evidence. If overdue maintenance cannot be explained or supported, the answer is 'no'.

3.6 Are monthly running hour reports sent to the company and are these reports consistent?

The inspector should verify that the vessel's engineers prepare and send monthly running hours reports to the technical managers. These reports should at least list the major components of the engines and their running hours since the last overhaul. It is recommended that this list also states the required maintenance intervals as specified by the makers, to easily identify if running hours have been exceeded. The inspector should verify that the hours stated in the report are consistent and properly following overhauls conducted. If no running hour reports are prepared, if these reports are not being sent to the technical managers, or if the reports are not completed in a correct and consistent manner, the answer should be 'no'.

3.7 Is maintenance properly recorded and is performed maintenance easy to trace back?

The inspector should verify that the vessel keeps properly documented maintenance records. The records should record maintenance jobs per equipment and allow for the maintenance history to be shown. It will not be sufficient for conducted maintenance to be described briefly, for example by stating 'OK' or 'Job done'; there should be a minimum description of what maintenance has been done. If proper maintenance records are not kept, the answer should be 'no'.

3.8 Is the main engine subjected to monthly performance testing?

The inspector should verify if the performance of the main engine is checked at least monthly and results properly recorded. If main engine performance checks are not performed and/or results not properly recorded on a monthly basis, the answer should be 'no'.

3.9 Do the performance records of the main engine suggest proper operation?

The inspector should review the running parameters (temperatures, pressures etc) of the last couple of main engine performance records to verify the main engine's performance. If possible, the parameters are compared against sea or shop trial results, if these are available on board. It is encouraged that the parameters are also compared to parameters recorded in the log book. If parameters show areas of concern, with temperatures or pressures being outside normal values or with deviations which cannot be explained or are not followed up, the answer is 'no'.

3.10 Have crankshaft deflections of the main engine been taken at the required intervals and were results satisfactory?

The inspector should verify that crankshaft deflections for the main engine have been taken at the intervals required by the makers. The deflection records should show values which are within limits. If not, it should be explained which corrective action is taken or

scheduled. If no crankshaft deflections are present or if there is insufficient follow-up given to the deviating values, the answer should be 'no'.

3.11 Are samples of the main engine lubrication oil taken at the required intervals, including weekly on-board samples?

The inspector should verify that samples of the main engine lubrication oil have been taken at intervals required by the makers and sent to a laboratory for analyses. In addition it is required that weekly on-board samples are taken and analysed to ascertain the BN, %water and viscosity. If samples are not taken at the required interval or not analysed, the answer should be 'no'.

3.12 Are recommendations (following) from the main engine lubrication oil analyses followed up?

The inspector should verify that recommendations and alerts made in the main engine lubrication oil analyses reports, are properly followed up. If not, the answer should be 'no'.

3.13 Are scavenging space inspections done at the required intervals and are findings documented?

The inspector should verify that scavenging space inspections for the purpose of ascertaining the condition of the piston, piston rings and liners, are carried out at intervals required by the makers. If not, the answer should be 'no'. If the vessel is equipped with a 4-stroke engine without scavenging ports, the answer is 'N/A'. In that case it should however be verified that piston and liner inspections are conducted in accordance with the intervals as per relevant question.

3.14 Are the scavenging space drains in good working condition?

The inspector should verify that the main engine scavenging space drains are open to allow proper draining of water. The surveyor should verify with the crew that regular draining is performed. If the scavenging drains are not working correctly, the answer should be 'no'.

3.15 Is there an adequate cylinder lubrication management system in place?

The inspector should verify that proper cylinder lubrication management is in place for vessels with 2-stroke engines. This system should include management of storage quantities, but should also specify cylinder lubrication settings to match circumstances such as new or worn piston rings. If no cylinder lubrication management is being executed, the answer should be 'no'. If the vessel is not fitted with a 2-stroke engine, the answer should be 'N/A'.

3.16 Is the stock (RoB) of lubrication oil for the main engine lubricating system sufficient for at least 1 filling of the sump tank?

The inspector should consult the engine details to ascertain the capacity of the main engine sump and the type of oil used. Accordingly, he should check that the onboard

stock of that oil is sufficient for at least one filling of the main engine sump tank. If the stock is less than the volume for filling the main engine sump tank, the answer should be 'no'.

3.17 Is the main engine sump tank lub oil level adequate for safe operation?

The minimum volume of lub oil in the main engine sump tank shall be kept to the equivalent of 1.3 L/kW with LO pump(s) in operation, or at 90% of the sump tank capacity with LO pumps stopped. The engine power, at original MCR on delivery, should be used as reference. The inspector should verify that the required quantity of lub oil in the main engine sump tank is established and an adequate lub oil quantity is kept. If the sump lub oil level is not at an adequate level, the answer should be 'no'.

3.18 Are the auxiliary engines subjected to regular performance testing?

The inspector should verify that the performance of the auxiliary engines is checked at least every two months or every 1,000 running hours, whichever comes first. If the auxiliary engine performance checks are not performed on a two-monthly basis or performed every 1,000 running hours, the answer should be 'no'.

3.19 Does the performance records of the auxiliary engine suggest proper operation?

The inspector should review the running parameters of the last couple of auxiliary engine performance test records to verify the engine's performance. If possible the parameters are compared against sea or shop trial result, if these are available on board. It is encouraged that the parameters are also compared to parameters recorded in the log book. If parameters show area(s) of concern, with temperatures or pressures which are being outside normal values and/or deviations that cannot be explained or are not followed up, the answer is 'no'.

3.20 Are the auxiliary engines subjected to monthly performance testing, which shows that they can reach 70% MCR?

If the auxiliary engines are operated at low loads for extended periods of time, fouling may occur due to which these engines may no longer be capable of producing their maximum designed load. To a certain level, reduction of power is inevitable because of normal wear and tear, but a certain minimum is required in order to also allow engines to be out of service.

The Chief Engineer shall provide records to the attending inspector showing performance tests of each auxiliary engine over the past 4 months. Through the records, the Chief Engineer shall demonstrate that each engine is subjected to a performance test, monthly, at electrical loads greater than 70% of its rated electrical load. Occasional testing below 70% MCR may be acceptable, provided that in general and over a longer period of time, testing is performed above 70%. Therefore, it may be necessary to check performance tests of various previous months.

If engines are not tested monthly or if records fail to show that the auxiliary engines are capable of running at minimum 70% MCR or if the records show inconsistencies /

deviations, the answer should be 'no'.

If no (monthly) performance tests are being conducted or if tests are done at lower loads, the answer should be 'no'.

For vessels exceeding 14,000 TEU capacity: For these vessels the Chief Engineer shall provide records to the attending inspector showing performance tests of each auxiliary engine over the past 4 months. Through the records, the Chief Engineer shall demonstrate that each engine is subjected to a performance test, monthly, at the following electrical loads:

- AE #1 and AE #4 - electrical load to be greater than 50% of its rated electrical load;
- AE #2 and AE #3 - electrical load to be greater than 60% of its rated electrical load.

If engines are not tested monthly or if records fail to show that the auxiliary engines are capable of running at the above loads or if the records show inconsistencies / deviations, the answer should be 'no'.

3.21 Are samples of the auxiliary engine lubrication oil taken at the required intervals, including weekly on-board samples?

The inspector should verify that samples of the auxiliary engine lubrication oil have been taken at intervals required by the makers and sent to a laboratory for analyses. In addition it is required that weekly on-board samples are taken and analysed to ascertain the BN, %water and viscosity. If samples are not taken at the required intervals or not analysed, the answer should be 'no'.

3.22 Are recommendations (following) from the auxiliary engines' lubrication oil analyses followed up?

The inspector should verify that recommendations and alerts made in the auxiliary engine lubrication oil analyses reports, are properly followed up. If not, the answer should be 'no'.

3.23 Is there a proper system in place for reporting of machinery defects to the company?

The inspector should verify if there is a proper system in place that allows reporting of defects related to the machinery to the company. If no (adequate) system is in place, the answer should be 'no'.

3.24 Is there adequate follow-up received from the company on defects?

The inspector should verify that adequate response and follow-up is given to defects reported. This includes technical consultation on possible solutions, feedback from makers and the provision of spare parts. If no such system is in place, the answer should be 'no'.

3.25 Are bunker procedures available and in use?

The inspector should verify that bunker procedures are available and are being used during bunker operations. Bunker procedures should include a pre-bunkering plan and a bunker checklist showing actions to be taken, such as setting up a bunker plan, applying scupper plugs, establishing communication etc. Bunker documentation

should show agreements with the bunker barge on maximum pressures, bunker rate etc. During bunkering these procedures should be displayed also at the bunker station. If no proper bunker procedures are available or not properly followed, the answer should be 'no'.

3.26 Is there a bunker checklist completed prior to, during and after bunkering?

The inspector should verify that bunker checklists are properly and truthfully completed prior to, during and after bunkering. If no checklist are completed, the answer should be 'no'. If the vessel's procedures do not include checklists, the answer may be 'N/A', provided that the bunker plan does include a proper preparation of bunkering.

3.27 Are bunker checklists counter signed by the bunker barge representative?

The inspector should verify that at least agreements on bunker rates and bunker pressures are properly communicated with the bunker barge representative and documented. Most commonly a bunker checklist is used to mutually agree on proper bunker preparation and agreement on bunker rates, which checklist should then be signed by both the Chief Engineer as well as the bunker barge representative. If no checklist is used, or alternatively no agreement is made on pumping rates, the answer should be 'no'.

3.28 Is the ECA change-over procedure in place and documented?

Vessels sailing within ECA's (Emission Control Areas), as laid down in Annex VI of the MARPOL rules, are committed to emitting maximum quantities of SO_x and NO_x. For example, the maximum sulphur content in fuel oils being used within ECA's is set at 0.1% as from 2015. Outside ECA's the current limit of sulphur is 3.5%, but will be reduced to 0.5% in 2020. The inspector should verify if the vessel has proper procedures in place to switch over to low sulphur fuels, sufficiently in advance of entering an ECA.

It must further be considered that there may be local emission control areas, not officially referred to as ECA, that also require low sulphur fuels to be used.

If change-over procedures are not available or not followed, the answer should be 'no'. If the vessel is not currently trading within ECA areas or other emission-controlled areas, the answer should be 'N/A'.

3.29 Is the NO_x record book of engine parameters filled in up to date?

In order to monitor engine emission, engine parameters for ships of which the keel is laid on or after 1 January 2000 are listed in the Technical File. The NO_x record book of engine parameters should display all changes to the parameters that can influence the emission of NO_x, such as changing the fuel injector type. The inspector should check the NO_x record book to see whether changes have been properly recorded. The NO_x record can be a separate book or it can be included in the technical file. More often administrations allow vessels to maintain official records in an electronic format.

If the NO_x record book of engine parameters is not available (on board ships of which the keel is laid on or after 1 January 2000) or if changes have not been properly

recorded, the answer should be 'no'.

If the vessel's keel is laid prior to 1 January 2000 and the vessel does not have a NOx record book, the answer should be 'N/A'.

3.30 Is the level of spare parts adequate?

The inspector should verify that the vessel maintains an apparently adequate stock of spare parts. It is not required to make a full inventory of all spare parts. Instead a general inspection of how spares are stored, packed and identified may be carried out to get an impression of the spare part management. Random checks against any computer-based spare part management system are recommended. If the inspector is of the opinion that spare parts are insufficient, not properly stocked or not maintained in accordance with computer-based quantities, the answer should be 'no'.

3.31 Can it be demonstrated that spare parts are OEM?

The inspector should, by random checks, verify that spare parts for the engines are OEM (Original Engine Manufacturer), thus being original parts. If parts are not original / OEM, the answer should be 'no'.

3.32 Have Critical Equipment and Critical Spares been identified and is a minimum stock of critical spares available? (For CMA Ships vessels - ENGINE-520 Critical Equipment risk matrix)

The inspector should verify that the crew is aware on Critical Equipment and Critical Spares (Minimum Spare Parts). The last inventory of Critical Spares shall not be older than 1 month and should show that the minimum stock is present on board.

For CMA Ships vessels, the inspector should refer to the ENGINE-520 Critical Equipment risk matrix. For third-party vessels, the inspector should use the critical spare list as per the ship manager ISM.

If Critical Equipment is not identified or if minimum stock of Critical Spares is not kept the answer should be 'no'.

3.33 Is there adequate follow-up received from the company on requirements for spare parts?

The inspector should verify that the vessel receives adequate follow up from the technical managers on a request for spare parts. It should be verified that spare parts are delivered without (extensive) delays. If there are long delays, reason should be provided with a sensible explanation, supported by documentation. If spare parts are not delivered on board when requested within an acceptable period of time or without unexplained long delays, the answer should be 'no'.

3.34 Is there an adequate system in place for defining roles and responsibilities of the crew within the engine room?

The inspector should verify that within the engine room there is a clear definition of the different roles of the crew and their responsibilities. For example, if the third engineer is responsible for the auxiliary engines, his tasks in this respect should be listed and explained and acknowledged by the third engineer. If no clear definition of roles is

present, the answer should be 'no'.

3.35 Is the engine room log book filled in properly and completely?

The inspector should check if the engine room log book is filled in properly and completely. Entries that are amended should be marked with initials. The use of correction fluid is never allowed as the log book is an official document. If the log book is not filled in properly, the answer should be 'no'.

3.36 Is the Oil Record Book filled in properly and completely?

The inspector should check if the Oil Record Book (ORB) is filled in properly and completely. Entries that are amended should be marked with initials. The use of correction fluid is never allowed as the ORB is an official document. More often administrations allow vessels to maintain official records in an electronic format, which includes the Oil Record Book. If the log book or ORB is not filled in properly, the answer should be 'no'.

3.37 Are fuel oil analyses conducted and results awaited prior to commencement of consumption?

The inspector should verify that the vessel has a system in place that arranges for fuel oil samples to be taken by the vessel during bunkering and these samples to be sent to a laboratory for verification of compliance with (at least) ISO8217 specifications. The inspector should verify that consumption of the fuel oil is not started before the analysis results are sent back to the vessel. If no bunker sample analysis system is in place, or if bunker consumption start before analysis results are received, the answer is 'no'.

3.38 Is cleaning of fuel tanks incorporated in the engine room PMS and performed according the set intervals?

The inspector should verify that bunker tanks (including settling and service tanks) are cleaned within intervals incorporated in the PMS. It should further be ascertained that appropriate preparations are being taken to comply with the international sulphur cap requirements, which may require tanks currently still containing high sulphur fuels, to be scheduled for cleaning. If fuel tank cleaning is not being conducted in accordance with requirements, the answer should be 'no'.

3.39 Is the condition of cooling water measured and is the cooling water treated?

The inspector should verify that the ship's engineers conduct regular testing of the cooling water, to confirm the water to have the appropriate chemical composition. Records should show that the dosage of chemicals is being adjusted on basis of the cooling water testing. If no cooling water testing is being done, or if no action is taken following deviating values, the answer should be 'no'.

3.40 Is the condition of boiler water measured and is the boiler water treated?

The inspector should verify that the ship's engineers conduct regular testing of the boiler water, to confirm the water to have the appropriate chemical composition. Records should show that the dosage of chemicals is adjusted on basis of the boiler water testing. If no cooling water testing is being done, or if no action is taken following deviating values, the answer should be 'no'.

3.41 Are there records available showing daily sounding of engine room tanks?

The inspector should verify that quantities in the different engine room tanks are ascertained daily. These quantities can be ascertained by manual soundings, level gauges or remote sounding systems. In case remote sounding systems are used, the inspector should check if manual soundings are regularly conducted to verify the correct readings of the remote system. If tank quantities are not being recorded daily or if remote tank quantity systems are not verified, the answer should be 'no'.

3.42 Are operation parameters of the engine room machinery stored (digitally or printed)?

The inspector should verify that the running parameters of the engine room equipment (pressures, temperatures) are recorded, either in the engine room monitoring system or printed on paper. If a printer is used, it is to be verified that the paper cannot jam or at least jamming is checked regularly, and that sufficient spare paper and ink is present. If these parameters are not being recorded, the answer should be 'no'.

3.43 Are engine room alarms stored (digitally or printed)?

The inspector should verify that the alarms of the engine room equipment (pressures, temperatures) are recorded, either in the engine room monitoring system or printed on paper. In case a printer is used, it is to be verified that the paper cannot jam or at least jamming is checked regularly, and that sufficient spare paper and ink is present. If these alarms are not being recorded, the answer should be 'no'.

3.44 Is there a proper system in place for controlling engine maker's service letters?

The inspector should verify that a system is in place to store and trace back all relevant service letters issued by engine makers. If no system to control receipt and documenting of the service letters is present, the answer should be 'no'.

3.45 Is the BWMS and associated equipment in good condition (filters, pumps, back flushing equipment)?

The inspector should perform a visual check of the overall condition of the BWMS and associated equipment. If the BWMS and associated equipment are not in good condition, the answer should be 'no'.

4. General

4.1 Are fire flaps tight and without excessive corrosion?

The inspector should verify whether fire flaps are tight and ventilation arrangements are free of excessive corrosion. If the fire flaps or dampers are not tight or show excessive corrosion, the answer should be 'no'.

4.2 Are all ventilation arrangements on the outside decks tight and without excessive corrosion?

The inspector should verify the whether ventilation arrangements, such as the vent heads are tight and free of excessive corrosion. If the ventilation arrangements are not tight or show excessive corrosion, the answer should be 'no'.

4.3 Are save-alls (outside the engine room) free of oil?

All save-alls, in particular those fitted on the main deck around fuel tank ventilation pipes and mooring winches, should be clean, of sufficient capacity and without free oil present. Furthermore, drain plugs must be available. If no save-alls are fitted where they should be or if save-alls are not free of oil, or drain plugs are not available, the answer should be 'no'.

4.4 Are ladders and accesses throughout the vessel safe?

The inspector should verify if ladders, accesses and entrances, for example to lashing platforms and into the holds, are safe. Ladders should be free of excessive corrosion and properly fixed to the ship's structures. Furthermore, safety cages around ladders, if fitted, should be free of wastage. Hold entrances should be clear and properly lit. If ladders, accesses or entrances are unsafe, the answer should be 'no'.

4.5 Are arrangements in place to prevent oily substances to be spilled overboard?

The inspector should verify that, where applicable, appropriate spill retainers are in place. This involves spill retaining edges on decks, but also the presence of scupper plugs for save-alls and deck scuppers. If no spill retention is present or if the spill retention is defective, the answer should be 'no'.

4.6 Is a SOPEP kit available on board, well stocked and readily available?

The inspector should verify that appropriate SOPEP spill cleaning material is present on board. This material should be readily available and properly stored at a designated SOPEP locker / storage place. The quantity of material should be sufficient and in compliance with the SOPEP inventory list. If SOPEP material is not properly stored or insufficient, the answer should be 'no'.

4.7 Are accommodation ladders, gangways and pilot ladders in a sound condition?

(Pilot) Ladders, rope ladders and gangways, should be safe means to board the vessel. The gangway should be sound, free of damage or bending and properly rigged with solid hand railings in place and a safety net applied. Steps should be checked for cracks and wire sheaves for excessive corrosion. The pilot ladder should be certified and without defects or unacceptable repairs. Furthermore, the inspector should verify if boarding can be done in a safe way without being endangered by ongoing cargo operations. If accommodation ladders, gangways or pilot ladders are not in a sound condition, the answer should be 'no'.

4.8 Are (portable) safety railings in a sound condition and in use along locations where people can fall down from (great) heights, such as open holds?

When there is a chance of people falling from (great) heights, sufficient fall protection should be installed. As per labour conventions, heights are defined as a minimum of 1.5 metre. Accordingly, lashing bridges and hatch traverses should be provided with safety railings, such as at, but not limited to, the outer ends. Also along open holds arrangements are to be made to prevent people falling into the hold. It should be verified that the fixed or portable railing (usually stanchions/sceptres, interconnected with ropes at two different levels) are in position and in a safe / sound condition. At places where no operations are taking place, such as for example often on the 20' positions outboard lashing platforms, railing is obviously not needed at that specific time. If, in way of open holds and at hatch traverses, no safety railing is fitted / erected, or if the railing is not in a safe condition, the answer should be 'no'.

5. LSA / FFE

5.1 Are the lifeboat(s), in sound condition and properly stored / secured?

The inspector should verify that the lifeboat(s) are in apparent sound condition and ready to launch. Launching equipment should be properly maintained and well-greased. Records should show that the boat has been launched (into the water) within the required intervals, which is usually every 3 months (see also question 1.41). Random checks of the inventory should indicate that this inventory is sufficiently stocked and maintained up to date. If the lifeboats are not in apparent sound condition, the answer should be 'no'.

5.2 TEST Can the lifeboat engines be started?

At least one of the lifeboat engines is to be started to confirm that it will run. If the engine will not start or does not run properly, the answer should be 'no'.

5.3 Are the life rafts in sound condition and are inspection dates valid / marked?

The inspector should verify that the life rafts are in an apparent sound condition, stating on them the last inspection date and validity. The inspector should also check that the hydrostatics release is properly connected and valid. If the life raft is not apparently sound or the hydrostatic release is not properly connected or not valid, the answer should be 'no'.

5.4 Is the MOB boat, if fitted, in sound condition?

The inspector should verify that the MOB / rescue boat is in an apparent sound condition and ready to launch. Launching equipment should be properly maintained and well-greased. Records should show that the boat has been launched within the required intervals, which is usually every 3 months. Random checks of the inventory should indicate that this inventory is sufficiently stocked and maintained up to date. If the MOB / rescue boat is not in an apparent sound condition or not regularly launched, the answer should be 'no'. If no separate MOB boat is fitted, this question should be 'N/A'.

5.5 Is all firefighting equipment valid, in sound condition and properly stored?

The inspector should verify that all fixed and portable firefighting equipment is in an apparent sound condition. This includes the CO2 system (bottles and release), fire hoses and nozzles and all portable fire extinguishers. The CO2 system and portable extinguishers should be labelled stating the last inspection and expiry date. If the firefighting equipment is not in an apparent sound and operational condition or not properly labelled, the answer should be 'no'.

5.6 Are sufficient immersion suits available?

The inspector should verify that a sufficient number of immersion suits is provided in compliance with the safety plan. If the number of suits is insufficient, the answer should be 'no'.

5.7 Is the crew wearing suitable PPE (Personal Protection Equipment)?

The inspector should verify that crew members on board are wearing PPE (Personal Protection Equipment) adequate for their working environment. This includes, but is not limited to, helmets, safety shoes, reflecting vests, gloves, goggles, ear protection and safety harnesses. The nature of the required equipment is depending on the working area and tasks being performed. If crew members are seen not to be wearing PPE adequate for their tasks or working area, the answer should be 'no'.

5.8 TEST Can the emergency fire pump be started?

The inspector should ask the accompanying crew member to start the emergency fire pump and connect two fire hoses to each utmost end of the fire line. It should be verified that the pump provides sufficient pressure, with the hoses producing a firm beam of approx. 20 m. outreach. If the emergency fire pump does not start or delivers insufficient pressure, the answer should be 'no'.

5.9 TEST Are engine room remote stops operational?

The inspector should ask for a few random engine room remote stops, for example for the engine room ventilation, to be activated. If the remote stops are not operational, the answer should be 'no'.

5.10 TEST Are remote closing valves operational?

The inspector should ask for at least one of the quick closing valves to be activated. Please note that only valves are tested that do not interfere with the vessel's operations. If the quick closing valve fails operation upon testing, the answer should be 'no'.

5.11 Is a safety plan provided in a water tight container near the gangway entrance

The inspector should verify the presence of a safety plan in a water tight receptacle near the accommodation entrance in way of the gangway at either side of the vessel. If this is not present, the answer should be 'no'.

5.12 Are muster lists posted and up to date?

The inspector should verify that muster lists are conspicuously posted at the designated areas, describing the contingency plans and responsibilities of all crew members. If no muster lists are posted or not stating tasks and responsibilities, the answer should be 'no'.

5.13 Are fire and smoke detection systems in a sound condition?

The inspector should verify the ship's (maintenance) records to see if the fire detection system is regularly tested. Most commonly detectors are tested every 3 months, where on a weekly basis a group of detectors is tested. It is encouraged that this group consist detectors of different loops. If records are not available or show that the fire alarm system is not tested regularly, the answer should be 'no'.

6. Accommodation

6.1 Are accommodation spaces free of increased fire risk (i.e. electrical)?

The inspector should verify that spaces inside the accommodation are free of any increased fire risk. For example, it should be verified, at random, that there are no unsafe (temporary) electrical arrangements or that crew members are not smoking at locations where this is not allowed. If an increased risk of fire is identified, the answer should be 'no'.

6.2 Is lighting within the accommodation working?

The inspector should verify that sufficient lighting is provided within the accommodation spaces. An occasional light being defective is not an immediate problem, but accessible spaces should be provided with sufficient lighting to allow safe passage and it should be clear that lights which are not working are being repaired. Lighting marked as emergency lighting should be operational at all times. If various lights are not working, or if one or more emergency lights are inoperative, the answer should be 'no'.

6.3 Is the accommodation in a clean and hygienic condition?

The inspector should verify that the accommodation, crew cabins and crew living quarters are well maintained and clean. Some random crew cabins need to be entered; these should be sufficiently equipped without major shortcomings to the upholstery. If the accommodation is dirty, unhygienic or with many items of upholstery broken, the answer should be 'no'.

6.4 Is the galley in a clean and hygienic condition?

The inspector should verify that the galley is well maintained and clean. Attention should be given to drain areas and cooking uptakes, which should be clear and clean, without excessive grease present. Adequate fire protection, such as an extinguisher and a fire blanket, should be readily available and not obstructed. If the galley, including drains and uptakes, is not clean, the answer should be 'no'.

6.5 Are the cold stores and food stores in clean and hygienic condition?

The inspector should verify that the cold stores and food stores are well organized and clean. Attention should be given to the cleanliness of hidden/concealed areas where dirt may collect. In addition the lock-in devices or lock-in alarms need to be operational; random testing may be advisable. If the cold stores are not properly organized, not clean, or lock-in devices not operational, the answer should be 'no'.

6.6 Are "locked in" alarms provided in the cold stores or can doors be opened from the inside?

The inspector should verify if the cold store doors can be opened from the inside or, if

not, an alarm button is provided and is working. If no emergency escape and/or alarm is present or if these means are not working, the answer should be 'no'.

6.7 Are records available showing that temperatures of the cold stores are monitored?

The inspector should verify that the temperatures of the cold stores are properly maintained and recorded. If the cold stores' temperatures are not maintained or recorded, the answer should be 'no'.

6.8 Are escape routes and relevant markings adequately posted?

The inspector should verify that throughout the accommodation spaces, emergency escape routes are properly marked using IMO type reflecting symbols. If these routes are not properly marked, the answer should be 'no'.

6.9 Is the hospital operational and are medicines well stocked / within expiry dates?

The inspector should inspect the hospital to verify it is apparently well organized and clean. The hospital should be well furnished with relevant equipment, such as a stretcher, oxygen resuscitation and a complete stock of medicines. The inspector should check, at random, if medicines are not expired and verify if the quantities comply with the inventory list. If the hospital is not properly organized, not clean, equipment defective or medicines absent or expired, the answer is 'no'.

6.10 Is garbage properly segregated and disposed of as per MARPOL rules?

The inspector should verify that waste is properly segregated and disposed of in accordance with MARPOL rules. If garbage is not segregated or properly disposed of, the answer should be 'no'.

7. Bridge

7.1 Is all bridge equipment in an apparent working condition?

The inspector should check for any signs that bridge equipment is not operational, such as posted notes, pieces of tape applied to disable the use of equipment, loose wires etc. If the operational condition is doubted, an operational test may be requested. Inspection of the bridge equipment should include inspection of the magnetic compass, to verify that it contains sufficient liquid and is free of air (bubbles), and whether a deviation curve or table is available/posted. If bridge equipment is not operational, the answer should be 'no'.

7.2 Are proper Pilot-Master information exchanges made and recorded during arrival and departure?

The inspector should verify that when sailing under pilotage, a proper Pilot-Master information exchange is executed. This information is usually included on a Pilot Card and should include details of the vessel, type of cargo on board, maximum draft of the vessel, trim, freeboard, air-draft, squat information, manoeuvring characteristics, type of engine(s), speed of engines at various power levels, type and number of propellers, number of starts available, time required for the vessel to go from full ahead to full astern, the number and position of the thrusters. This exchange is to be done prior to the Pilot providing sailing/manoeuvring instructions and the exchange is to be properly recorded. If no Pilot-Master exchange is carried out, not completely executed or if it is not recorded, the answer should be 'no'.

7.3 Are the vessel's manoeuvring characteristics clearly displayed?

The inspector should verify that the vessel's main manoeuvring characteristics are clearly displayed on the bridge, for example on a 'wheelhouse poster'. These characteristics should include minimum stopping distance, turning circles and various propulsion details. If this information is not available, the answer should be 'no'.

7.4 Are emergency instructions clearly displayed?

The inspector should verify that emergency steering instructions are clearly indicated on the bridge. These instructions should show which actions the bridge officer needs to take to switch over to emergency steering. It should further be clear to the officers which ways to communicate with the steering gear room are available for emergency steering. If no emergency steering instructions are posted or communication not in place, the answer is 'no'.

7.5 Does the company have a procedure indicating the minimum UKC (Under Keel Clearance)?

It should be verified if the vessel's management company has indicated what minimum UKC (Under Keel Clearance) should be maintained. Commonly, UKC requirements are divided into open and confined waters. If the management company has no guidance

set out for the UKC, the answer should be 'no'.

7.6 Is passage planning performed correctly, including UKC, and from berth to berth?

Passage planning should be conducted for each and every voyage, no matter how short these voyages, or no matter how frequently that voyage is made. In no way should the crew rely on information from previous voyages to be still accurate. Furthermore passage planning should always include sailing under pilotage (berth to berth). The passage planning should include the minimum UKC (Under Keel Clearance) for each leg of the voyage. This UKC should be calculated by using chart depth, tidal information as well as a correction for squat. The inspector should check records of voyage planning of some randomly selected previous trips and verify whether passage planning is performed correctly. If no passage planning is conducted or performed incompletely or does not include a proper UKC calculation, the answer should be 'no'.

7.7 Are Master and deck officers familiar with the operation of the ECDIS system fitted on board?

The Master and deck officers should be familiar with the operation of the ECDIS. Master and deck officers, randomly selected, should be able to demonstrate the operation of the ECDIS and that ECDIS is corrected up to date. If the officers questioned are not able to demonstrate the operation of the ECDIS system, or if the system is not up to date, the answer should be 'no'.

7.8 Is the vessel's BNWAS system used?

The inspector is to verify that during navigation the BNWAS (Bridge Navigation Watch Alarm System) is activated and operational. It should be verified that the control keys for deactivating the system are controlled, for example being in the possession of the Master only. If the BNWAS system is not used, not operational or if deactivation is not protected, the answer should be 'no'.

7.9 Are Master's standing orders and night orders communicated / displayed?

The inspector should verify the presence of a document or booklet with the Master's standing and night orders. If these orders are not present and available to the watch keeping officers, the answer should be 'no'.

7.10 Have all navigation officers signed for Master's (standing) orders?

The inspector should verify that standing orders are signed by the watch keeping officers. If these orders are not signed for, the answer should be 'no'.

7.11 Are weather and navigation warnings received?

The inspector should verify that all means for receiving weather updates and navigational warnings are fully operational, such as weather faxes, NAVEX etc. This may

include the use of weather routing (software), if and when the vessel is instructed to use this. If weather messages and navigation warnings are not being received, the answer should be 'no'.

7.12 Are compass deviations properly displayed?

The inspector should verify the availability of a compass deviation curve / table and that the Magnetic Compass and Gyro compass errors are regularly determined by different means and registered. If compass errors are not being ascertained or recorded, the answer is 'no'.

8. Hull and deck

8.1 Is the outside hull free of excessive corrosion?

The inspector should inspect the outside hull for excessive corrosion. A distinction should be made between corrosion that is severe (darkly coloured) and superficial /progressing corrosion, commonly of a light brown colour. Furthermore, some allowance can be made for situations where only a relatively small area is affected by corrosion. The extent of corrosion should be considered excessive if there is severe corrosion over a reasonable area or when there is superficial corrosion all over a large area. If the hull is displaying excessive corrosion, the answer should be 'no'.

8.2 Are hull markings (name, logo, draught marks) clear and legible?

All markings on the hull, such as the vessel's name, company logos and in particular Plimsol and draught marks, should be properly legible. In case hull markings are worn and not properly legible, the answer should be 'no'.

8.3 Are the decks free of excessive corrosion?

The inspector should check whether the main decks are free of excessive corrosion or wastage. If corrosion is present but remains to be limited in extent, or superficial in nature, this may be acceptable but should be mentioned in the summary of the report, such as to alert awareness. If the deck and deck fitted structures are displaying excessive corrosion, or wastage, the answer is 'no'. Breakdown of paint that is purely cosmetic, as well as slight corrosion, should be mentioned in the survey summary but is not a reason for answering this question with 'no'.

8.4 Are deck fittings (piping, sounding pipes, lashing gear storage) free of excessive corrosion and properly identified?

The inspector should check whether deck structures and fittings, such as deck posts, coaming stanchions, hand rails, lashing gear storage bins, lighting fixtures etc. are free of excessive corrosion or wastage. If corrosion is present but remains to be limited in extent or superficial in nature, this may be acceptable but should be mentioned in the summary of the report, such as to alert awareness. If the deck and deck fitted structures are displaying excessive corrosion, or wastage, the answer is 'no'. Breakdown of paint that is purely cosmetic, as well as slight corrosion, should be mentioned in the survey summary but is not a reason for answering this question with 'no'.

8.5 Is the deck lighting in working condition?

The inspector should (as by random tests) verify that the deck lights are in working order. An occasional light being defective is not an immediate problem, but accessible spaces should be provided with sufficient lighting to allow safe passage and it should be clear that lights which are not working are being repaired. Lighting marked as emergency lighting should be operational at all times. If various lights are not working,

or if one or more emergency lights are inoperative, the answer should be 'no'.

8.6 Are the hatch covers / hatch pontoons free of excessive corrosion?

The inspector should check if hatch covers or hatch pontoons are free of excessive corrosion that could affect their integrity and / or ability to be correctly positioned. If corrosion is present but remains to be limited in extent or superficial in nature, this may be acceptable but should be mentioned in the summary of the report, such as to alert awareness. If the hatches (hatch pontoons) are displaying excessive corrosion or wastage, the answer should be 'no'.

8.7 Can hatches / hatch pontoons be properly closed and secured?

For vessels with hydraulically or electrically operated hatches, the operational condition should be verified (hydraulic cylinders, wheels, hinges etc.). It should be checked whether side cleats are in a good condition (without excessive corrosion, have sound rubber washers and nuts that can be turned by hand). For vessels with hatch pontoons that are placed in position by cranes, it should be verified that guiding and securing arrangements are in a sound condition. It should be noted that vessels with lashing bridges are not required to have pontoon securing devices. In case hatches or hatch pontoons are not in a sound condition or cannot be properly secured, the answer is 'no'.

8.8 Is the hatch coaming free of excessive corrosion?

The inspector should check if the hatch coamings/coaming stays are free of excessive corrosion. If corrosion is present but remains to be limited in extent or superficial in nature, this may be acceptable but should be mentioned in the summary of the report, such as to alert awareness.

8.9 Are container sockets, hatch lifting points and lashing rings free of excessive corrosion and/or wastage?

The inspector should, through random inspection at sufficient scattered locations, verify whether container sockets, lashing eyes (also known as lashing plates or lashing rings) and hatch cover lifting points are free of excessive corrosion and/or wastage. If corrosion is present but remains to be limited in extent or superficial in nature, this may be acceptable (but should be mentioned in the summary of the report, such as to alert awareness) If sockets and lashing points are displaying excessive corrosion and/or wastage, the answer should be 'no'.

8.10 Is the lashing equipment in an apparent sound condition?

The inspector is requested to inspect, at random, the lashing gear and twist locks in use and/or located in storage bins. It should be verified that the equipment is in apparent operational condition and free of excessive corrosion or wastage. It should be verified that no mixed (left-right closing) twist locks are used since this presents a high risk of confusion during cargo operations, whereby containers may remain unsecured after loading or alternatively remain locked during discharge. For semi-automatic twist locks

it should be verified that no single-wire twist locks are mixed with dual-wire types. In case conventional type twist locks are used, for example as bottom twist locks, it should be verified whether no right-hand and left-hand locking twist locks are used simultaneously. If the lashing equipment is not apparently operational, in a poor condition or of mixed types, the answer should be 'no'.

8.11 Is the lashing performed properly and in accordance with the CSM?

It should be verified that the lashing material is applied in accordance with the cargo securing manual (CSM). The CSM specifies which type of material is to be used (twist locks, lashing rods) and the locations where this material is to be applied. Furthermore, the inspector should check if lashings to "in transit cargo" is applied correctly. In a longitudinal direction the lashing rods should be as vertical as possible, the turnbuckle securing pins should be provided in such a way that the locking pin cannot dislocate and lashings should be properly tightened. If the lashing is not applied properly or not in accordance with the CSM, the answer should be 'no'.

8.12 Are the hatch traverses, lashing bridges and lashing platforms in a sound condition?

Hatch traverses on container ships are normally arranged with walking platforms or gratings to facilitate application of lashing material. Often also lashing bridges are in place. Particular interest should be given to the outer lashing platforms, where lashing teams are working close to the side of the vessel. The inspector should verify if the walking platforms, gratings, accesses and ladders are free of excessive wastage. Limited deformation as part of normal wear and tear, is acceptable. If hatch traverses and/or lashing platforms are not in a good condition the answer should be 'no'.

8.13 Are hydraulic lines and hoses free of leakage and in a sound condition?

If hydraulic arrangements are present on the (main) deck, for example for hatch cover operations or operation of mooring equipment, it should be verified that lines and hoses of this system are sound and free of leakage by random inspection. Piping and couplings should be checked for excessive corrosion, hoses should be checked for wear and should be free of excessive twisting and not painted over. If the hydraulic pipes and hoses are leaking and/or apparently corroded or worn, aged, twisted or painted over, the answer should be 'no'. If there are no hydraulic systems on outside decks, the answer should be 'N/A'.

8.14 Are reefer sockets in an apparent sound condition?

It should be verified, by random inspection, that the reefer sockets are properly mounted, apparently operational and well accessible. Furthermore, the sockets should be properly protected from seawater ingress by applying caps when the sockets are not in use. In case extension cables or distribution boxes (commonly also referred to as 'splitter boxes') are used, these should be without damage and positioned so that they are not easily damaged by lashing gear being dropped. Furthermore, extension cables should be mounted clear from the deck and properly shielded from seawater. If reefer

sockets, cables and distribution boxes are loose, damaged and/or apparently out of order, the answer should be 'no'.

8.15 Are mooring winches and (anchor) windlasses without excessive wear?

The inspector should check the mooring and anchoring arrangements on forecastle and poop deck. Anchor windlasses and mooring winches should be checked for worn brake lining. Capstans and hawseholes should be inspected to verify that rollers can rotate freely. If anchoring and mooring arrangements are not satisfactory, the answer should be 'no'.

8.16 Are mooring lines in a sound condition and properly secured?

Mooring lines should be properly secured to keep the vessel safely alongside. This not only involves proper securing to winches and bollards, but includes regular monitoring by the crew to ensure that lines are under sufficient tension.

Mooring lines should not be secured on warping drums (drum ends). Where mooring lines are secured on split drums (often referred to as (auto-)tension drum), this drum should not contain more than one layer of mooring line, as more layers may become entangled. Mooring winches accommodating spring lines should never be put on auto-tension mode. For winches that are on manual mode, it should be verified that the clutches are disengaged.

Mooring ropes should be checked for excessive wear, for example shown by broken strands and/or fraying. Splices are allowed, as long as they are of sufficient length.

The vessel should maintain a mooring log, keep certificates of the mooring lines at hand and preferably record when mooring lines are taken in use.

If mooring lines are not properly secured, are in worn condition, or if documentation relating to the mooring is not available, the answer should be 'no'.

8.17 Is the forecastle store tidy and free of traces of water ingress?

The inspector should verify that the forecastle store is tidy and all items are properly secured. Furthermore, the inspector should check for traces of water ingress, both from the deck into the forecastle store, as well as into the bow thruster compartment. If the forecastle store is untidy or shows traces of water ingress, the answer should be 'no'.

8.18 Is the bow thruster motor in an apparent sound condition?

The inspector should verify that the bow thruster is in a sound condition without any leakage of oil or water. The inspector should check the electric motor, the sealing onto the thruster tunnel and bilge in the bow thruster room. If the bow thruster is not in a sound condition or shows leakage, the answer should be 'no'.

8.19 Are the hold bulkheads without excessive corrosion?

The inspector is requested to obtain a representative picture of the condition of the holds and the bulkhead structures. It should be verified if bulkheads are without

excessive corrosion or even wastage. Obviously, a poor cosmetic appearance is of lesser concern and accordingly some rust formation can be allowed. If the hold bulkheads have excessive corrosion or wastage, the answer should be 'no'.

8.20 Are the hold cell guides without excessive corrosion and damage which can adversely affect cargo operations?

The inspector is requested to obtain a representative picture of the condition of the cell guides. Cell guides always have corrosion from the contact with the containers, so rust is inevitable. It should however be verified that the cell guide supports / bulkheads are without excessive corrosion or even wastage, and whether they are free of damage adversely affecting cargo operations. If the cell guide supports / hold bulkheads have excessive corrosion or wastage, the answer should be 'no'.

8.21 Is the hold tank top and are the tank top stacking cones / container guides without excessive corrosion?

The inspector should, by random inspection of (empty) holds or empty stacks, verify the condition of the tank top and container securing arrangements, such as container sockets and/or (bottom) stacking cones. The tank top should be without excessive corrosion affecting the structure and has to be reasonably clean, being without debris / objects that could compromise bilge suction, such as bottles, rags, ropes etc. Flush tank top sockets should be clear of dirt and stacking cones should be without wastage. If the tank top or container cones are corroded, or if the hold or flush sockets are not clean, the answer is 'no'.

8.22 Are the hold ventilation arrangements in a sound condition?

Holds of container vessels are usually fitted with mechanical ventilation, which could be of either a supply or an exhaust type. Container ships certified to carry dangerous cargo underdeck, must, according to SOLAS, be fitted with mechanical ventilation fans that can be put in the exhaust mode.

Cargo holds designated for carrying refrigerated containers may be fitted with additional ventilation fans which operate in the supply mode, but different lay-outs exist. Often also using the natural vent openings are fitted, such as on the side of the hatch pontoons.

In most cases the air is distributed throughout the holds, using a system of ventilation ducts.

The inspector is to verify that this ventilation system is in a good condition, with the ventilation ducts sound and without excessive corrosion. Vent flaps, both inside the ducts as well as fire flaps on the outside of the hold, should be operable and sound.

Fans do not need to be tested during survey, but if they are running, the inspector is asked to randomly verify that the fans do not make any abnormal sounds, which may be a sign of bearing failure.

If vent ducts are damaged, heavily corroded, if flaps are not operable or if fans have abnormal sounds when running, the answer should be 'no'.

8.23 Are the hold bilges clean and clear?

The inspector should verify that the hold bilges are clear of significant debris or dirt that could negatively affect the bilge suction. If the bilge is not clean and clear, the answer should be 'no'. If no hold bilges are accessible during inspection, the answer should be 'N/I'.

8.24 TEST Are the hold bilge alarms operational?

The inspector is requested to activate at least one bilge alarm and record that the alarm is activated on the corresponding alarm panel. If the hold bilge alarm is not operational, the answer should be 'no'. If no hold bilges are accessible during inspection, the answer should be 'N/I'.

8.25 Are hold ladders and walkways in a sound and accessible condition?

The inspector should verify that all ladders and walking platforms in the holds are safe to access without major defects or excessive corrosion. If the ladders or platforms are not in a safe condition, the answer should be 'no'.

8.26 Is the bunker manifold apparently sound, save-alls clean and empty, and are save-all drain plugs available?

The inspector should verify that the bunker manifolds are apparently operational. This involves manifolds for fuel oil, lubrication oil and sludge disposal. The connections should be without excessive corrosion. Flanges fitted should be tight with packings apparently sound and bolts/nuts free to move. Gauges fitted and drains for taking samples should be operational. If the bunker manifold is not in an apparently operational condition, the answer should be 'no'.

8.27 Are deck fitted lifting appliances in sound condition (bunker hose crane, store crane)?

The inspector should check the visual condition of the deck fitted lifting appliances that not used for cargo operations, such as store cranes, bunker hose cranes etc. The inspector should verify that these lifting appliances are visually sound, apparently operational and that the (runner) wires are in good order and properly greased. This question does not deal with cargo gear / cranes, which are dealt with in separate questions. In case any of these lifting appliances is in use during survey, proper operation should be verified. If the cranes and /or the wires are not in sound condition, the answer should be 'no'.

9. Cargo gear

9.1 Are the cargo crane pedestals free of excessive corrosion?

If the vessel is fitted with cargo gear, the inspector should visually inspect the pedestals of the cranes for damage and/or excessive corrosion. Some superficial (spot) corrosion may be acceptable, but more excessive corrosion should be arrested to avoid structures becoming affected. Accordingly, if the paint layer is worn, revealing corrosion at multiple locations and/or over a larger area, the answer should be 'no'. Furthermore, if the pedestals have damage, cracks or deformations of which it cannot be excluded that it may affect this structure now or in the future, the answer should also be 'no'. If the vessel is not fitted with cargo gear, the answer should be 'N/A'.

9.2 Are the cargo crane houses / turrets free of excessive corrosion?

If the vessel is fitted with cargo gear, the inspector should visually inspect the crane houses for damage and/or excessive corrosion. Some superficial (spot) corrosion may be acceptable, but more excessive corrosion should be arrested to avoid structures becoming affected. Accordingly, if the paint layer is worn, revealing rust at multiple locations and/or over a larger area, the answer should be 'no'. Furthermore, if the crane houses have damage, cracks or deformations of which it cannot be excluded that it may affect this structure now or in the future, the answer should also be 'no'. If the vessel is not fitted with cargo gear, the answer should be 'N/A'.

9.3 Are the cargo crane jibs free of excessive corrosion?

If the vessel is fitted with cargo gear, the inspector should visually inspect the cargo crane jibs and jib fittings for damage and excessive corrosion. Some superficial (spot) corrosion may be acceptable, but more excessive corrosion should be arrested to avoid structures becoming affected. Accordingly, if the paint layer is worn, revealing corrosion at multiple locations and/or over a larger area, the answer should be 'no'. Furthermore, if the cargo crane jibs show damage, cracks or deformations of which it cannot be excluded that it may affect this structure now or in the future, the answer should also be 'no'. If the vessel is not fitted with cargo gear, the answer should be 'N/A'.

9.4 Is the interior of the cranes safe to access (clean, unobstructed, not slippery)?

The inspector should verify that the (commonly used) pedestal entrances and intermediate decks are free of stored materials which obstruct the safe passage and are clean of oil (not slippery). These areas normally serve as preferred exit also in case of emergencies and accordingly blockage is to be avoided. If the crane entrance and intermediate decks contain materials that block passage or are slippery, the answer should be 'no'.

9.5 Is the crane slewing gear sound?

The inspector should verify that the slewing ring and slewing gear are free of excessive corrosion and are well-greased (not over-greased). If the slewing ring gear / teeth are displaying corrosion and/or if there is insufficient, or too much, grease applied, the answer should be 'no'.

9.6 Is the crane machinery without apparent defects?

The inspector should verify that the crane machinery is free of defects affecting the operation. This could include excessive corrosion, parts broken off or removed, temporary repairs etc. If the crane machinery is not in apparent sound condition, the answer should be 'no'.

9.7 Is the crane machinery free of leakage?

Most cargo cranes have hydraulic systems in place for operation of the crane. For most cranes these systems operate the hoisting, luffing and slewing of the crane. Also cranes where the main operation is driven by electric motors, may have small hydraulic systems for ancillary machinery. Oil drops on machinery and / or hydraulic hoses, oily decks or the presence of free oil in save-alls is a sign that there is leakage of hydraulic oil. If leakage of oil is present, either inside or outside the crane, the answer should be 'no'. In case the crane does not have any hydraulic systems, the answer should be 'N/A'.

9.8 Are crane wires without corrosion and sufficiently greased?

All cargo cranes have a hoisting wire. Luffing may also be arranged through wire operation, but can also be arranged by hydraulic cylinders. The condition of a wire is not always easy to assess because the core of the wire is not visible. Wires that have been mounted for a (too) long time may have become corroded or may have broken strands. If over a length of 8 times the rope diameter the number of broken strands is more than 10% of the total number of wires, the wire rope should be discarded. Also deformations, such as kinking, crushing, loosening of strands, looping or bird caging, are a reason for discard. In order to preserve the condition of the wire as much as possible, it should be properly greased, in particular the parts that are (always) exposed and those parts that have the most contact with the sheaves. If a wire is damaged, kinked, corroded, or not properly greased, the answer should be 'no'. Furthermore, a wire is connected with one dead end on the wire drum, the other connected to a socket on the crane house or jib. If those sockets are damaged, improperly fitted or excessively worn, the answer should be 'no'.

9.9 Are all wire sheaves on the crane operational and free of excessive corrosion?

The wire sheaves, both on the crane house and on the jib, should properly guide the wire in its movement and accordingly the sheaves should be sound with their bearings properly greased. If the sheaves fail to allow a smooth motion of the wires, the wires may become damaged. If the sheaves are damaged or excessively corroded, the answer should be 'no'.

9.10 Are the crane hook and block operational?

It should be ascertained that the hook is in sound condition and that wear resulting from contact with (steel) is not excessive. Very often the hook is installed on a swivel, that allows unrestricted rotation of the hook. Most commonly hooks have safety latches installed, that should prevent slings / chains from jumping off the hook. If the hook is excessively worn, the swivel unable to rotate or the safety latches inoperable, the answer should be 'no'.

9.11 Is the SWL clearly marked on the jib of the crane or on their crane housing?

It should be clear to all working with a lifting device, including to people hooking up a load, that the crane is restricted in its hoisting capacity. This limit is indicated by the safe working load (SWL), being a metric tonnage against a maximum outreach, and is usually marked on the jib of the crane. If the SWL, including maximum outreach, is not marked on the outside of the crane or if the marking is no longer distinct, the answer should be 'no'.

10. Engine room

10.1 Is the engine room displaying proper housekeeping?

The inspector should check if the engine room is tidy and that no unsecured objects are present. Furthermore, the engine room should be clean, without oil present in the bilges or inside the save-alls and without oily rags lying around. If the engine room is not tidy or if it is dirty, the answer should be 'no'.

10.2 Is the main engine in an apparent sound condition and without leaks?

The main engine should be visually inspected for its apparent sound condition. It is requested that the inspector verifies the operation with the engine in service and verify, as much as possible, that it is running in order. Running parameters as recorded in the log book or obtained during the inspector's presence, should be investigated and possibly compared with shop and/or sea trial results. The inspector should check if the outside engine casing is free of (excessive) leakage of fuel oil, lubricating oil or water. An engine is a machine so the presence of slight traces of oil may be accepted, however if there is excessive leakage which is obviously ignored by the engineers, this is not acceptable. Furthermore, it should be checked if gauges are legible and whether any temporary solutions for faulty equipment are present, which may jeopardise the engine's continuous operations. If the main engine is not in apparent sound condition and/or has excessive leakage the answer should be 'no'.

10.3 Can the engineers show that they are familiar with local main engine manoeuvring?

The attending inspector should ask the engineer to show that they are aware of how local (emergency) manoeuvring of the main engine is executed. The engineers should be able to show how to switch over, how to start and control the main engine and how communication with the bridge is made. If the engineers are unable to demonstrate local main engine manoeuvring, the answer should be 'no'.

10.4 Are the auxiliary engines in an apparent sound condition?

The auxiliary engines should be inspected for their apparent sound condition. It is requested that the inspector verifies the operation with the engines in service and verify, as much as possible, that they are running in order. Running parameters as recorded in the log book or obtained during the inspector's presence, should be investigated and possibly compared with shop and/or sea trial results. For each engine the inspector should check if the external engines' casing is free of (excessive) leakage of fuel oil, lubricating oil or water. An engine is a machine so the presence of slight traces of oil may be accepted, however if there is excessive leakage which is obviously ignored by the engineers, this is not acceptable. Furthermore, it should be checked if gauges are legible and whether any temporary solutions for faulty equipment are present, which may jeopardise the risk to the engines' continuous operations. If the auxiliary engines are not in apparent sound condition and/or have excessive leakage the answer should be

'no'.

10.5 TEST Is the power management system for the generators working?

The inspector should verify that the power management system that controls the load division over the generators is working properly. A proper test depends on the lay-out of the system, must include verification of an even load division and automatic starting of a (second) generator when the first generator reaches a certain limit of load. If the power management system is not apparently properly operating, the answer should be 'no'.

10.6 TEST Are the fuel oil and lubrication oil treatment systems in an apparent operational condition?

The inspector should verify if the fuel oil and lubrication oil treatment systems, comprising of purifiers, heaters and (booster) pumps, appears functional and in sound condition. All purifiers should be tested in running condition. Performance testing includes verification of the correct temperature settings as per recommendation / analysis reports. Large quantities of spare parts lying around or apparent temporary solutions should be further investigated. If the fuel oil treatment plant is not apparently sound, if performance testing is not performed or if corrective action(s) on defects are not taken, the answer should be 'no'.

10.7 Is insulation in place around hot surfaces and is this insulation in good condition?

The inspector should verify that insulation is present around hot surfaces, such as fuel lines. The inspector should check that this insulation is in a sound condition and is not defiled with oil. If insulation is missing, damaged or defiled with oil, the answer should be 'no'.

10.8 Is the fuel oil leak arrangement for main engine high pressure fuel pipes without leaks and the leak off tank alarm tested?

Double skinned fuel pipes between the fuel pump and injector are used as a safety measure to protect against the risk of fire in the case of a fuel pipe failure. To be effective, they must be correctly fitted and sealed against external leakage at fuel pump and injector. In case of a leak, the fuel oil drains to a leak tank which should be fitted with an operational level alarm. It should be verified that level alarm testing is done regularly as part of the vessel's maintenance schedule.

If the fuel oil leak off arrangements for high pressure fuel pipes are defective or if the alarm of the leak tank is not regularly tested the answer should be 'no'.

10.9 Are exhaust piping for diesel engines, boilers and other hot surfaces in the vicinity of fuel and lub oil pipes protected against spray?

Surfaces with temperatures above 220C which may be affected as result of a leak from an oil or fuel systems shall be properly insulated.

The attending surveyor should verify that exhaust piping for diesel engines, boilers and other hot surfaces in the vicinity of fuel and lub oil pipes protected against spray

If exhaust piping for diesel engines, boilers and other hot surfaces in the vicinity of fuel

and lub oil pipes protected against spray, the answer should be 'no'.

10.10 Are switchboards free of significant earth faults (resistance >1 MΩ)?

The inspector should verify that the MSB / Main Switch Board (normally in way of the engine control room) and the ESB / Emergency Switchboard (normally in way of the emergency generator) are in apparent sound condition, without panels removed, wiring exposed or meters removed. The inspector should further verify that the boards are free of earth faults, which means that resistance should be minimum 1 MΩ (Class requirement).

If the minimum requirement of 1 MΩ is not achieved, or if there are any other deficiencies on the electrical switchboards, the answer should be 'no'.

10.11 Are instructions for the bilge water separator posted?

The inspector should check whether instructions for operating the bilge water separator (or oily water separator OWS), are conspicuously posted. These instructions should include notification of the use of overboard pumping through oil content monitoring equipment (such as 15 ppm). If instructions are incomplete or missing, the answer should be 'no'.

10.12 TEST Can the engineers show proper functioning of the bilge water separator including the three way valve?

The inspector should ask the attending engineer(s) to demonstrate that the bilge water separator has properly functioning OCM (oil content monitoring / 15 ppm). The OWS three way valve to overboard / recirculation must be checked for correct operation, based on the ppm result.

If the crew cannot demonstrate the functioning or if the OCM does not work, the answer should be 'no'.

10.13 Is the primary bilge tank operational and bilge holding tank clear?

Modern ships may have a primary bilge tank, which collects engine room bilge water and skims off most of the oil that may be in it. The tank is heated and has a cascade system, which should remove most of the oil and lead to fairly clean bilge water in the final stage, so that it can be transferred into the bilge holding tank. The inspector should further verify that the final stage of the primary bilge tank and the bilge holding tank do not contain excessive oil. If the bilge tanks are containing too excessive quantities of oil, the answer should be 'no'.

10.14 Are the boilers in good order and do they appear to be well maintained?

The inspector is to ascertain the visual condition of the boilers (oil fired and exhaust gas) and verify that the boilers are in apparent sound condition, without leakage of fuel or water. It is requested that the inspector verifies the operation of equipment in service and verify, as much as possible, that it is running in order. Furthermore, the safety valves should be apparently sound, not manipulated and without signs of leakages. Insulation

should be intact and properly applied. Where automated boilers are fitted, they should be operating in automatic mode. Boiler controls should not be overridden or by-passed. The boiler panel must be checked for cleanliness and to ensure that there are no electric circuit breaker wire jumps installed which are not as per design. In case of any observation the answer should be 'no'.

10.15 Is the sewage plant in good order and appears to be well maintained?

The inspector should verify the sewage installation is apparently sound, all components in place and without the tank or connected lines leaking. Valves should be free to move and the system should be free of temporary arrangements or by-passes. If the sewage plant is not in a sound condition, the answer should be 'no'.

10.16 Are save-alls in the engine room free of excessive quantities of oil?

The inspector should verify that save-alls fitted around fuel oil and lubricating oil equipment are free of excessive quantities of oil. If save-alls contain excessive amounts of fuel oil or lubrication oil, the answer should be 'no'.

10.17 Are engine room pumps in an apparent sound condition?

The inspector should verify that the pumps in the engine room are in sound condition without excessive leakage of fuel oil, lubrication oil, steam or water. If one or more pumps are defective or showing excessive leakage, the answer should be 'no'.

10.18 Is other engine room equipment, not dealt with elsewhere in this questionnaire, in a sound condition?

The inspector is to ascertain if other engine room equipment, not listed elsewhere in this questionnaire, is in a sound condition and apparently running in order. This would comprise air compressors, refrigeration compressors, heat exchangers, fresh water generators, etc. Condition of F.O. and L.O purifiers is included in question 10.6. If one or more of these auxiliary appliances are not apparently sound, the answer should be 'no'.

10.19 TEST Can the engine room ventilation dampers be closed?

The inspector should verify that the engine room ventilation dampers are without excessive corrosion, free to move and that these can be closed using emergency closure. If the ventilation dampers are not in a sound condition or cannot be properly closed, the answer should be 'no'.

10.20 Is the steering gear free of oil leakage?

Testing of the steering gear is not required, unless there is reason to doubt proper performance. The inspector should check if the steering gear is free of leakage of hydraulic oil. It is not uncommon that the area underneath the steering gear is slightly oily or may have some grease present, but it should be ascertained that the gear is free

from a continuous flow of oil, emerging from seals or pipe connections. If oil leakage is apparent, the answer should be 'no'.

10.21 Is the communication system between the bridge and the steering gear room functional?

The inspector should ascertain that a proper communication system between the steering gear room and the bridge is present and is working. Most often there is a common telephone system and a back-up system, such as a sound powered telephone, which are recommended to be tested. In case communication depends on portable radios, a proper reception is to be verified by a test. If no proper and adequate communication system is set up between the steering gear and the bridge, the answer should be 'no'.

10.22 Are objects in the steering gear room properly secured?

Objects (oil drums, chemicals etc.) in the steering gear room should be properly secured, to avoid these items blocking access to the steering gear in case of an emergency. If objects are not properly secured, the answer should be 'no'.

10.23 Is the emergency generator in an apparent sound condition and the fuel tank filled in with enough fuel?

It should be verified that the emergency generator appears sound, without any signs of leakage of fuel oil, lubricating oil or water. Furthermore, it should be evident that the emergency generator is tested regularly, for example by checking relevant (maintenance) records. If there are any doubts about the performance, a test may be requested.

The diesel oil tank for the emergency generator should always maintain a level which is capable of 18 hrs of uninterrupted operation.

If the emergency generator is not apparently operational or the fuel tank is not kept with enough fuel the answer should be 'no'.

10.24 TEST Can emergency power be demonstrated?

The inspector should ask for the emergency generator to be started or, in case of emergency battery power, to be shown that the system is able to provide back-up power. If no emergency power can be demonstrated, the answer should be 'no'.

10.25 Are the emergency batteries in an apparent fit/sound condition?

The inspector should verify that the emergency starting batteries for the emergency generator are in a sound condition, which includes verification there is no spillage of acid, no corrosion on the connections, cables in sound condition etc. If the emergency batteries are not in apparent sound condition, the answer should be 'no'.

10.26 Are starting instructions for the emergency generator posted?

The inspector should check if instructions for starting the emergency generator and

connecting it to the (emergency) switchboard, are conspicuously posted. If no instructions are present, the answer should be 'no'.

10.27 Is there evidence of safe working practices being applied in machinery spaces?

The inspector should verify that all work performed in the engine room and other machinery spaces is conducted according to common safety procedures that would allow a safe working environment to be created. This may include the use of additional safety equipment / PPE, installing fall protection, cleaning up spills on the job etc. If no proper safe working environment is created for executed jobs, the answer should be 'no'.

10.28 In case of UMS, is the engineer's alarm system operational and regularly tested?

If the engine room is approved for UMS (Unmanned Machine Spaces), it should be verified that engine alarms are properly relayed to remote alarm panels in the accommodation spaces. A few random panels must be inspected, which is obviously best witnessed when the engine room is already switched over to 'unmanned'. If the alarms are not properly relayed, the answer should be 'no'. If the vessels is not approved for UMS, the answer should be 'N/A'.

10.29 Are engine room lifting appliances in sound condition (overhead crane)?

The inspector should check the visual condition of the engine room fitted lifting appliances such as the overhead travelling crane. In case any of these lifting appliances is in use during survey, proper operation should be verified. If the cranes and /or the wires are not in sound condition, the answer should be 'no'.

10.30 Is the alarm panel in the engine control room free of disabled alarms?

The inspector should verify on the alarm panel in the engine control room if any alarm is disabled. If an alarm is disabled the answer should be 'no'. In the comment box the disabled alarms have to be mentioned.

10.31 Are the air dryers for the control air system in good working condition?

The inspector should verify that the air dryers for the control air system are in good working condition. The drains should be clean, of sufficient capacity, and working correctly. No water should be present in the airdryers. If the air dryers and/or drains are not working correctly, the answer should be "no".

10.32 Are the turbocharger filters clean (not fouled/clogged)?

The inspector should check if the filters of the main engine and auxiliary engines turbochargers are foul / clogged. If a filter is foul / clogged the answer should be 'no'.

10.33 Are the drains for the main engine and auxiliary engine turbochargers operational?

The inspector should verify that the turbocharger drains are in good working condition. The drains should be clean, of sufficient capacity and the drain valves should be

working correctly and easily to be operated. If the TC drains are not working correctly or cannot be operated, the answer should be "no".

10.34 Is the backflush of the automatic lubrication oil filter working correctly?

The inspector should verify that the automatic lubrication oil filter is apparently in a sound condition and the backflush filter is working correctly. The by-pass valve has to be in closed position. If the automatic lubrication oil filter is not in a sound condition, or the back flush filter is not working correctly or the by-pass valve is not closed, the answer should be 'no'.

10.35 Are the lubrication oil (LO) analysis carried out as per CMA Ships required intervals included in the PMS?

The inspector should verify that the lubrication oil analysis intervals are kept as per CMA Ships required intervals included in the PMS. If the intervals are not kept, the answer should be 'no'.

10.36 Are the LO/FO analysis (excluding LO analysis for ME and AE), in the past six months free of any alert, or if not, are corrective actions being taken?

The surveyor should check the results of recent LO (lube oil) and FO (fuel oil) analyses excluding LO analysis for the main engine and auxiliary engines that are dealt with in questions 3.12 and 3.22.

If any alert was advised on the analysis reports during the last six months, proper corrective action should have been taken. The corrective actions taken should have been recorded and detailed. If no sampling has been carried out or no corrective actions were taken, the answer should be 'no'.

10.37 Are the running hour records of the past six months showing that at least one auxiliary engine lubrication oil purifier is running 24/7?

The inspector should verify the running hour records of the machinery equipment and check that there has been always one lubrication oil purifier running, to ensure 24/7 lube oil purification. Records should be checked for the last 6 months. If no 24/7 purification was done for the last 6 months, the answer should be 'no'.

10.38 TEST The inspector shall perform an onboard lubrication oil analysis on the main engine sump to ascertain LO water content.

The inspector should perform an onboard lubrication oil test on the main engine sump lubrication oil with a ship's engineer to evaluate the water content. For testing the ship's onboard testing kit should be used. If a test cannot be performed due to lack of testing material, or water content above 0.3% is observed during the test, the answer should be 'no'.

10.39 TEST Are the fuel oil settling tanks regularly drained?

The inspector should verify that the fuel oil settling tanks are regularly drained by the crew. Also, the surveyor shall drain the water in the settling tank together with a crew member and verify that the water quantity is restricted. If the crew is not able to or

draining of the fuel oil tank(s) is not carried out, the answer should be 'no'.

10.40 Is the water in boiler cascade tank free of oil and sediments?

The inspector should verify that the water in the boiler cascade tank is free of oil and sediments. If the boiler cascade tank is not free of oil or sediments, the answer should be 'no'.

10.41 Can the auxiliary boiler operate with either on FO or DO?

The inspector should verify that the auxiliary boiler can operate with either diesel oil or fuel oil. It should be verified that the system shows control valves to switch over. If the auxiliary boiler can operate only with DO the answer should be 'no'.

10.42 Are the bilges in the engine room free of makeshift/additional piping?

The inspector should verify that the bilges in the engine room are free of "makeshift" piping or additional piping. If "makeshift" piping or additional piping is present on the bilge scuppers, the answer should be 'no'.

10.43 Is the stand-by function on all pumps on auxiliary engine / main engine fuel oil system working properly?

The inspector should verify if the stand-by function of all pumps on auxiliary engines and main engine fuel oil system are working properly. If the stand-by function of the pumps on auxiliary engines / main engine fuel oil system is not working correctly, the answer should be 'no'.

10.44 Is the stand-by function of all pumps on auxiliary engines / main engine lubrication oil system working properly?

The inspector should verify if the stand-by function of all pumps on auxiliary engines and main engine lubrication oil system are working properly. If the stand-by function of the pumps on auxiliary engines / main engine lubrication oil system is not working correctly, the answer should be 'no'.

10.45 Is the stand-by function of all pumps on auxiliary engines / main engine cooling water system working properly?

The inspector should verify if the stand-by function of all pumps on auxiliary engines and main engine cooling water system are working properly. If the stand-by function of the pumps on auxiliary engines / main engine cooling water system is not working correctly, the answer should be 'no'.

10.46 TEST Is it possible to switch between the primary and secondary electric 440/220 transformers?

The inspector should verify that switching between the primary and secondary 440/220 transformers is possible. If switching between transformers cannot be performed, the answer should be 'no'.

10.47 Are the main engine and auxiliary engines fuel oil viscometers in a good working condition and regulating in auto mode?

The inspector should verify that the main engine and auxiliary engines fuel oil viscometers are in a good working condition and regulating in auto-mode. If the viscometers are not in a good working condition or not regulating in auto mode, the answer should be 'no'.
